



Rincoin Monetary Policy and Security Scenario Analysis

Evaluation of Post-840k Monetary Schedules

Rincoin Community Forge

2026-08-07

 Discussion Draft

Document version 1.0 · Public-review date 2026-08-07

This analysis presents the complete declared scenario space before selecting a principal review set. No candidate is preferred or adopted, and this document does not change the rules enforced by deployed software.

Abstract

Rincoin targets one-minute blocks but inherited a 210,000-block binary-halving interval. The resulting subsidy cuts occur roughly every five months rather than roughly every four years, creating repeated 50% shocks to nominal miner revenue. This report compares the unchanged rule with bounded and unbounded alternatives for block height 840,000 using reproducible integer simulations, explicit policy gates, a compact unweighted monetary score, and a separately reported empirical security overlay.

The deterministic layer first defines scenarios S0 through S7 with variants and calculates all bounded schedules with integer base-unit arithmetic. Eligibility gates and five equally weighted criteria are then applied to every bounded candidate. Numerical scoring is one input to the later review-set selection, not an adoption decision or a complete measure of security.

Frozen chain, fee, market, and rental evidence is evaluated only after the deterministic results and selection method. It tests current fee-replacement, liquidity, rental, concentration, confirmation and implementation risks but does not convert observations into a causal forecast. Every later candidate conclusion remains conditional on source correction, arithmetic review, implementation tests, chain-separation decisions, service readiness and voluntary network adoption.

Executive Summary

Rincoin targets a block every 60 seconds and reduces the deployed subsidy by half every 210,000 blocks. A 210,000-block epoch therefore lasts about five target months, making Rincoin's 50% reward shocks approximately ten times more frequent in calendar time than the corresponding Bitcoin cadence. The decision at block height 840,000 is whether to leave that rule unchanged or replace it with a slower deterministic decline.

The Customized Halving work identified Rincoin's unusually frequent 50% subsidy shocks as a problem requiring explicit review (Tokino 2026). That diagnosis does not imply automatic acceptance of its exact parameters, governance model or implementation choices introduced in the Tokino's study - they are included here as one of several bounded candidate scenarios.

The evaluation uses a transparent decision tree:

1. If no correction is desired, S0 remains the deployed baseline.
2. If correction is desired but bounded issuance is not required, S3 and S7 remain policy comparators.
3. If both correction and bounded issuance are required, the bounded candidate set is S1, S2/a, S2/b, S4, S5/a, S5/b, S6/a, S6/b, and S6/c.

Five equally weighted monetary criteria compare the bounded candidates: reward-path correction, monetary discipline, ten-year scheduled supply expansion, nominal subsidy runway, and rule clarity and auditability. Rule clarity evaluates each proposal as originally published. Reconstruction, specification and vectors produced for this review do not retroactively improve that source score. Monetary totals, fixed-phase duration, transition shocks and the complete score table are presented before any principal set is selected.

For deterministic calculations, S6/a, S6/b and S6/c are fixed height-only interpretations of the three study scenarios. The S6/b review rule derives a finite 0.6 RIN phase ending at height 234,587,500 and exactly 168,000,000 RIN maximum

scheduled issuance. Underclaims do not extend the phase. The study's governance contingency is discussed but is not inserted into that consensus function. S7 remains the separate unbounded inspected code path.

The empirical overlay is deliberately separate from the monetary score. Frozen chain data test current transaction fees, halving windows, rental exposure, liquidity, operator concentration and confirmation assumptions. These data are thin, partly operator-supplied and unsuitable for causal claims. They inform later trade-off analysis without preselecting a monetary rule.

The document proceeds in dependency order: definitions and decision gates; every scenario and source; deterministic methods and results; complete scoring; transparent review-set selection; controlled and representative comparisons; empirical security evidence; implementation risks; and conclusions. No candidate is adopted or preferred in this front matter.

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1 Introduction

This report compares deterministic monetary schedules that can apply at and after block height 840,000. It separates historical documentation, deployed consensus, reproducible simulation, observed evidence, policy criteria, and implementation risk. It is not itself a consensus specification or a network decision.

The Customized Halving work identified Rincoin’s unusually frequent 50% subsidy shocks as a problem requiring explicit review ([Tokino 2026](#)). The study also identified a contradiction between the original whitepaper’s 168,000,000 RIN maximum-supply statement and the deployed binary-halving schedule, which approaches approximately 21 million RIN before integer-rounding effects. While this does not imply automatic acceptance of its exact parameters, governance model or implementation choices introduced in the Tokino’s study, its contribution to these findings is recognized and the scenarios proposed by the study are included in the candidate scenarios this analysis compares.

The comparison asks:

- Does the current 210,000-block binary-halving cadence require correction?
- Is bounded issuance a required policy property?
- What maximum issuance, near-term supply expansion, subsidy runway, and activation shock follow from each complete rule?
- Which conclusions are deterministic, observed, derived, or assumption-based?
- What evidence or implementation finding should change the later review-set selection?

2 Starting Points and Initial Assumptions

2.1 Document Hierarchy

Layer	Role
Rincoin whitepaper	Compact statement of Rincoin identity, consensus design and monetary rule
Scenario analysis	Evidence status, scenarios, simulations, economic reasoning, risk register and unresolved decisions (this document)
Consensus-change specification	Normative formulas, activation rules, integer arithmetic, block validation rules and test vectors
Rincoin Community Core implementation and tests	Executable consensus behaviour and release artifacts

2.2 Source Hierarchy

Claims use four evidence categories:

1. **Original-whitepaper claims** establish what the historical RinCoin document stated.
2. **Deployed source-code facts** establish current consensus behaviour as enforced by validating software.
3. **Chain-data and simulation-derived results** establish observed history, calculated supply paths and projected consequences under explicit assumptions.
4. **Policy/design recommendations** state proposed future choices for public review.

The original whitepaper confirms at the documentation level: BLAKE3 -> Argon2d -> SHA3-256, a 60-second target block interval, a 50 RIN initial subsidy, a stated maximum supply of 168,000,000 RIN, Dark Gravity Wave v3, no premine, no developer tax, CPU/GPU accessibility as a design claim, and ASIC resistance as a design claim ([RinCoin Whitepaper 2025](#)).

The deployed binary-halving schedule confirms all whitepaper claims except the 168,000,000 RIN maximum-supply statement. The deployed code does not implement the stated 168 million figure; before integer-rounding effects it approaches approximately 21 million RIN ([Rin-coin contributors 2026a](#), [2026c](#)). This is an objective documentation-versus-consensus inconsistency. The scenarios below test different policy responses to that inconsistency.

The deployed source code defines actual consensus behaviour that has been already established. Therefore, a documentation claim must not be converted into a deployed-consensus fact without a proper reconsideration.

2.3 Scope Boundary: Proof of Rinne

Tokino's broader paper presents **Proof of Rinne (PoR)** as a long-horizon research architecture in which a separate temporal layer would eventually re-circulate assets considered irretrievably lost or abandoned, while attempting to preserve a fixed macro-level supply bound and a continuing miner-security budget (Tokino 2026).

The idea is relevant to long-term research and should not be dismissed merely because it is incomplete. It addresses a real question: whether a capped proof-of-work currency can maintain incentives after ordinary issuance becomes negligible without adopting perpetual inflation. However, PoR is not evaluated as one of the present Layer-1 monetary scenarios because it depends on mechanisms that are outside the block-840,000 decision and are not yet specified at implementation level. In particular, the present materials do not provide a consensus-safe method to prove permanent abandonment rather than long-term inactivity, a complete ownership-recovery and adversarial threat model, deployed reference code, normative test vectors, or a near-term activation path.

Chapter 3 of Tokino v1.6.4 remains relevant here because it defines concrete Layer-1 Customized Halving schedules that can be simulated. The later PoR architecture is therefore acknowledged but excluded from scoring by scope, not rejected as an idea, but not treated as current Rincoin consensus.

2.4 Tail Emission Legitimacy

This analysis does not accept the 1.5% annual retention-loss assumption used in Tokino's study as a sufficient basis for introducing perpetual tail emission. The parameter is a model-calibration assumption rather than an independently validated empirical constant.

Available Bitcoin loss estimates either measure narrowly defined provably lost supply or infer a much broader probably lost supply through heuristics. They vary substantially by methodology and do not establish a stable, memoryless annual loss rate. Some more recent estimates are more consistent with losses having been concentrated disproportionately in Bitcoin's early years than with a constant annual hazard. Loss rates may also decline as asset values, custody technology, backup practices, and user awareness improve, although the true loss process cannot be observed directly from blockchain inactivity alone.

This does not imply that tail emission should be rejected permanently. Every bounded non-S0 scenario covered by this analysis keeps the block subsidy above 0.5 RIN for more than ten target years and non-zero for more than a century.

This provides substantial time for future maintainers and the community to evaluate improved estimates of coin loss, fee revenue, hashrate, mining economics, and observed security conditions.

If later evidence shows that a permanent subsidy is necessary, the monetary rules could be amended through a future explicit consensus change before the scheduled subsidy reaches zero. Introducing perpetual issuance now would therefore commit the protocol to an unbounded monetary path on the basis of a weakly supported assumption.

Postponing that decision preserves optionality and does not require the project to resolve today a zero-subsidy security question that lies far outside the current planning horizon. Conversely, an unnecessary perpetual emission, once embedded in expectations and mining economics, may be substantially harder to remove.

2.5 Continuity Concepts

For purpose of this analysis, three distinct continuity concepts are defined:

Ledger continuity means preservation of existing blockchain history and UTXO state. This is a non-negotiable condition - all valid pre-activation blocks, balances and ownership claims must remain valid for every scenario examined.

Consensus-code continuity means preservation of current validation rules without change. Any post-840,000 subsidy change is a hard fork and therefore does not preserve consensus-code continuity after activation, so no scenario can claim that property except the unchanged baseline scenario where no consensus rules change.

Monetary-intent continuity means alignment with an explicitly stated interpretation of the original whitepaper's monetary intent. Because the original whitepaper's documented 168,000,000 RIN maximum-supply statement conflicts with the deployed binary-halving schedule, no interpretation is neutral: this analysis compares treating that statement as a historical upper bound, a policy target, or a comparison reference. A historical documentation error is one possible interpretation, but the available evidence does not establish which parameter was erroneous. Therefore, this analysis compares the scenarios to both the original whitepaper's 168,000,000 RIN statement and the deployed binary-halving schedule. It also recognizes that the finite issuance has been an intention of both the original whitepaper and the deployed code, even if the exact upper bound is disputed.

2.6 Deterministic Supply Accounting

Consensus monetary rules evaluated in this report are deterministic functions of block height:

$$B_{\max}(h) = f(h),$$

where $B_{\max}(h)$ is the maximum permitted block subsidy at height h .

They must not depend on actual minted coinbase outputs, circulating supply, spendable supply, UTXO supply, underclaimed subsidy or transaction fees. Underclaimed subsidy is not reissued. Transaction fees do not count as newly issued supply.

The analysis distinguishes:

- **original 168,000,000 RIN statement:** a maximum-supply statement in the original whitepaper that conflicts with deployed binary-halving consensus;
- **effective maximum scheduled issuance:** the maximum amount a deterministic subsidy schedule can issue;
- **final spendable or circulating supply:** an economic or chain-observation quantity that may be lower than scheduled issuance.

2.7 RinHash and Difficulty Source Note

The deployed source references identify the RinHash construction and Argon2d parameters ([Rin-coin contributors 2026f](#)) and the DGWv3 implementation and selection ([Rin-coin contributors 2026b](#)).

RinHash is memory-hard and was designed to increase the cost of fixed-function specialization relative to simple hash-only proof-of-work algorithms. ASIC resistance and CPU/GPU accessibility are design goals and threat-model statements, not permanent empirical guarantees.

The currently deployed Core implementation has been verified as using Argon2d v1.3 with time cost `t_cost = 2`, memory cost `m_cost = 64` KiB / 64 one-kilobyte blocks, and `lanes = 1` ([Rin-coin contributors 2026f](#)). These parameters are not changed by any monetary schedule evaluated here. Current settings remain light enough that commercially available FPGA implementation might be plausible.

While this analysis recognizes that the RinHash design is a deliberate attempt to reduce the cost advantage of fixed-function specialization, it does not evaluate the security of RinHash or its resistance to future ASIC development. The analysis assumes that the current proof-of-work function remains unchanged for all scenarios, and any future changes to RinHash would require a separate consensus proposal, review, and testing process.

3 Evaluation Criteria

A monetary-policy comparison should not allow a scenario to win merely because it avoids change, nor should it reward the same property several times under different names.

Therefore, the analysis separates **eligibility checks** from **five scored criteria**. The checks identify scenarios that do not satisfy a declared policy requirement. The scores then compare the remaining trade-offs without hidden weights.

3.1 Eligibility Checks

Check	Pass condition	Consequence
Deterministic consensus rule	Subsidy is fully defined by block height and fixed integer constants.	A scenario that fails is not implementation-ready and is excluded from candidate scoring.
Corrective relevance	The rule materially addresses the repeated 50% subsidy cuts produced by 210,000-block epochs at a 60-second block interval.	S0 remains the indispensable baseline, but it is not scored as a corrective candidate once the rapid-halving problem statement is accepted.
Bounded issuance	Maximum scheduled issuance is finite and predetermined.	S3 and S7 remain useful security-economics comparators, but they are excluded from candidate scoring under the fixed-supply requirement.

The first check is technical. The latter two are explicit policy gates adopted for this comparison. Raw deterministic results remain available for every scenario, including the excluded baselines and comparators, but only scenarios passing all three checks enter the candidate score table.

3.2 Scored Criteria

Scores use a 0-4 scale:

- **4 points:** strong fit with the criterion;

- **3 points:** good fit with a clear trade-off;
- **2 points:** mixed result;
- **1 point:** weak fit;
- **0 points:** fails the criterion.

Criterion	Question asked	Scoring rule
Reward-path correction	How well does the rule correct the rapid 50% reduction cadence caused by 210,000-block epochs at a 60-second block interval?	4 = recurring reductions are no greater than 10%; 3 = 50% reductions remain but are spaced roughly four years apart; 2 = several manual 50% steps remain within the first 15 years; 1 = only limited mitigation; 0 = unchanged 50% reductions every 210,000 blocks.
Monetary discipline	Is issuance finite and scarce without being dominated by a cap-filling constant phase?	4 = finite, no constant phase, and final issuance no more than one third of 168M; 3 = finite, no dominant constant phase, and final issuance between one and two thirds of 168M; 2 = finite above two thirds of 168M without a dominant constant phase; 1 = finite, but more than half of final issuance occurs in a constant phase; 0 = unbounded.
Ten-year scheduled supply expansion	How much new issuance is scheduled during the first ten years after activation, relative to the 19,687,500 RIN scheduled before height 840,000?	4 = no more than 25%; 3 = more than 25% and no more than 60%; 2 = more than 60% and no more than 100%; 1 = more than 100% and no more than 150%; 0 = more than 150%.
Nominal subsidy runway	How long does the maximum block subsidy remain at or above 1 RIN?	4 = at least 25 years or indefinitely; 3 = 10 to less than 25 years; 2 = 5 to less than 10 years; 1 = 2 to less than 5 years; 0 = less than 2 years. This is only a nominal runway proxy, not a complete security-budget model.
Rule clarity and auditability	Can the full schedule be explained and independently reproduced without hidden state or many exceptions?	4 = one familiar formula or parameter change, with no special terminal phase; 3 = one formula plus one floor or tail, or one unusual coefficient requiring derivation; 2 = several fixed phases, or one formula plus a special terminal height; 1 = several phases plus a cap-filling terminal boundary; 0 = state-dependent or incompletely specified.

The five scores deliberately cover different questions: the problem being corrected, the monetary result, scheduled supply expansion, the nominal mining runway, and the clarity of the consensus rule. Scheduled supply expansion is a deterministic measure; actual economic dilution also depends on active supply, lost coins, market demand, price, and liquidity. The scores are not a substitute for price, hashrate, fee-market, or miner-cost analysis.

Rule clarity and auditability evaluates the proposal as originally published, including the precision of its rule, the derivation of its parameters, and unresolved governance or terminal behavior. Later reconstruction, specification, vector generation, or implementation work prepared for this review does not retroactively increase the source proposal's score. This rule applies equally to every externally sourced scenario.

The arithmetic total is a compact balance indicator only. It must be read together with the eligibility checks and the underlying deterministic values.

4 Scenario Definitions

In this chapter, the scenarios are defined and the rationale for each is explained. The scenarios are numbered S0 through S7, with variants where appropriate. Each scenario is described in terms of its subsidy rule, its expected behavior, and its relevance to the overall analysis.

Unless stated otherwise, scenarios use $1 \text{ RIN} = 10^8 = 100,000,000$ base units, preserve all blocks below height 840,000, and define alternatives from the block height 840,000 (or fifth reward epoch the way epochs are defined by the current consensus rules) onward. In this document, “post-840k” means block height 840,000 and later.

The descriptions below define the rules and explains why each case is included. They do not select a winner. Comparative advantages and disadvantages are assessed after the generated results.

4.1 Overview of Scenario Strategies and Objectives

The alternative scenarios examined in this analysis explore three related issues:

- **Rapid subsidy decline** — under the deployed schedule, the block subsidy is halved every 210,000 blocks, or approximately every five target months. This both large or unusually frequent.
- **The 21M/168M documentation–consensus inconsistency** — the original whitepaper documents an issuance figure of 168 million RIN, while the deployed consensus rules imply maximum scheduled issuance of approximately 21 million RIN.
- **The long-term mining security budget** — under a finite subsidy schedule, the block subsidy eventually reaches zero, leaving transaction fees as the only remaining block-reward component. It is uncertain whether fee revenue alone would provide a sufficient long-term incentive to secure the network.

These issues are interdependent. A rule that addresses one of them may improve or worsen another. The scenario set is therefore designed to expose the trade-offs among subsidy cadence, total scheduled issuance, and the long-term subsidy path.

The classifications below are independent of one another. Each alternative scenario belongs to one category under each issue: its subsidy-reduction strategy, its treatment of the documented 168 million RIN figure, and its terminal-subsidy behaviour.

Note: S0 is the deployed baseline and is therefore not classified under the three issues.

4.1.1 Subsidy-reduction strategy

For the first issue, the scenarios use three principal approaches:

- **Reduction-factor scenarios (S1, S2/a, S2/b, S3, S4)** — these scenarios retain the existing epoch length but replace the 50% reduction with a different multiplicative factor applied at each epoch boundary.
- **Extended-epoch scenarios (S5/a, S5/b)** — these scenarios retain 50% subsidy-reduction steps but increase the interval between later reductions.
- **Customized Halving¹ scenarios and the related implementation path (S6/a, S6/b, S6/c, S7)** — these scenarios use manually specified, piecewise-defined block-height boundaries and subsidy levels. S6/a, S6/b, and S6/c are study scenarios, while S7 represents the separately inspected implementation path derived from the same broader design direction.

4.1.2 Treatment of the documented 168 million RIN figure

For the second issue, the scenarios fall into three groups according to their maximum scheduled issuance:

- **Below 168 million RIN (S1, S2/a, S2/b, S5/a, S5/b)** — these scenarios treat 168 million RIN as an upper bound rather than a target that must be reached. They increase maximum scheduled issuance relative to S0 but remain below the documented figure.
- **Exactly 168 million RIN (S4, S6/a, S6/b, S6/c)** — these scenarios treat 168 million RIN as a terminal maximum-scheduled-issuance target.
- **Unbounded issuance (S3, S7)** — these scenarios do not enforce 168 million RIN as a finite Layer-1 cap. Their scheduled issuance eventually exceeds that figure and continues without a finite terminal maximum.

4.1.3 Terminal-subsidy behaviour

For the third issue, the scenarios use three approaches:

- **Continued decline to zero without a fixed terminal phase (S1, S4, S5/a, S5/b)** — these scenarios continue reducing the subsidy until it reaches zero under the applicable integer-arithmetic rule. They do not provide a perpetual subsidy.

¹The term “Customized Halving” is used in Tokino’s study to describe scenarios based on manually specified, piecewise subsidy schedules. Although these schedules do not always involve literal halving, this analysis retains the term for consistency with the source material.

- **Finite fixed-subsidy phase (S2/a, S2/b, S6/a, S6/b, S6/c)** — at a defined height, these scenarios switch to a constant subsidy that remains in effect until the scenario’s terminal issuance limit is reached. This fixed phase may be substantially longer than the preceding declining phase and, in some cases, may last for centuries.
- **Perpetual fixed subsidy (S3, S7)** — at a defined height, these scenarios switch to a constant non-zero subsidy that continues indefinitely. Their maximum scheduled issuance is therefore unbounded.

4.1.4 Scenario Strategies Summary

Table 4: Summary of scenario strategies and their treatment of the three principal issues.

Scenario	Subsidy strategy	168M treatment	Terminal behaviour
S0	None (baseline)	Below 168M	Declines to zero
S1	Reduction factor	Below 168M	Declines to zero
S2/a	Reduction factor	Below 168M	Finite fixed phase
S2/b	Reduction factor	Below 168M	Finite fixed phase
S3	Reduction factor	No finite cap	Perpetual fixed subsidy
S4	Reduction factor	Exact 168M target	Declines to zero
S5/a	Extended epochs	Below 168M	Declines to zero
S5/b	Extended epochs	Below 168M	Declines to zero
S6/a	Customized path	Exact 168M target	Finite fixed phase
S6/b	Customized path	Exact 168M target	Finite fixed phase
S6/c	Customized path	Exact 168M target	Finite fixed phase
S7	Customized path	No finite cap	Perpetual fixed subsidy

4.2 S0 - Unchanged Binary Halving

Scenario S0 is the baseline for all other scenarios because it follows the subsidy rule that has governed the chain since genesis.

The deployed binary-halving logic continues unchanged. Half of the subsidy is subtracted every 210,000 blocks until integer right-shift rounding reaches zero. In base units, the rule for epoch e is:

$$b_1 = 5,000,000,000 = 50 \text{ RIN} \quad b_e = \left\lfloor \left(1 - \frac{1}{2}\right) b_{e-1} \right\rfloor, \quad e = 2, 3, 4 \dots$$

S0 requires no post-840k consensus change. It is included to show the deployed-code baseline against which all hard-fork alternatives must justify their additional issuance and implementation risk.

4.3 S1 - One-Twentieth Reduction Without a Floor

Beginning at block height 840,000, subsidy follows a recursive 19/20 rule per 210,000-block epoch instead of binary halving. This is nominally a 5% reduction, with integer-floor rounding after each epoch; effective percentage reductions can differ slightly after rounding.

The one twentieth in S1 is chosen as a reduction factor ten times smaller than the original one half, matching the fact that Rincoin's target block interval is ten times shorter than Bitcoin's. The relationship is approximate rather than exact: after 10 epochs, the subsidy factor is $(1 - \frac{1}{20})^{10} \approx 0.5987$, not 0.5.

S1 is the first from the reduction-factor scenarios, proposing a direct way to slow down the fast subsidy decline caused by 60-second blocks while retaining a simple geometric rule by changing the factor used for the new subsidy calculation only. It preserves deterministic height-only accounting based on geometric function, uses the same epoch length and avoids dynamic supply tracking.

There is no fixed floor and no constant terminal phase in S1. Integer base-unit rounding eventually brings subsidy to zero. In base units, the post-840k rule is:

$$b_4 = 625,000,000 \quad b_e = \left\lfloor \left(1 - \frac{1}{20}\right) b_{e-1} \right\rfloor, \quad e = 5, 6, 7 \dots$$

The first post-840k subsidy for the fifth epoch is 5.93750000 RIN.

4.4 S2 - One-Twentieth Reduction With Finite Constant-Phase Variants

S2 variants apply the S1 one-twentieth reduction until the next calculated subsidy would fall below a selected fixed phase amount. They then use that fixed subsidy for a predetermined number of blocks and stop before exceeding the 168,000,000 RIN ceiling.

These variants test the policy choice of using most of the historical 168 million upper bound while keeping Layer-1 issuance finite and height-deterministic. They also expose how long a constant phase must last when it is used to approach that ceiling.

4.4.1 S2/a - Finite 0.5 RIN Phase

S2/a uses one-twentieth reductions until the next epoch subsidy would fall below 0.5 RIN. It then uses 0.5 RIN for a finite predetermined block range and terminates.

4.4.2 S2/b - Finite 1 RIN Phase

S2/b uses one-twentieth reductions until the next epoch subsidy would fall below 1 RIN. It then uses 1 RIN for a finite predetermined block range and terminates.

4.5 S3 - Perpetual 0.5 RIN Tail-Emission

S3 applies the S1 one-twentieth reduction until 0.5 RIN and then keeps a perpetual 0.5 RIN subsidy, even after the 168,000,000 RIN ceiling is reached. The perpetual phase is a tail-emission design choice that can be argued as a direct miner-security-budget mechanism. It is included to show the effect of a permanent nominal subsidy on scheduled issuance and to provide a comparator for bounded schedules.

This scenario isolates the explicit-tail-emission design choice. It is included because a permanent nominal subsidy can be argued as a direct miner-security-budget mechanism, but it must be compared separately from finite schedules.

4.6 S4 - Idealized 113/114 Cap-Derived Geometric Schedule

S4 uses an epoch coefficient of 113/114 and has no fixed subsidy phase. The coefficient has been defined as mathematically closest to the original whitepaper's declared 168,000,000 RIN maximum when applied as a geometric post-840k rule under integer base-unit arithmetic.

This scenario is a cap-derived comparator. It tests a smooth geometric approach to the historical upper bound without selecting a fixed terminal subsidy.

4.7 S5 - Extended-Epoch Binary-Halving Variants

S5 tries a different approach to slowing the rapid 50% reductions. It retains the original binary-halving rule but lengthens the epoch from 210,000 blocks ten times to 2,100,000 blocks to match the bitcoin halving interval.

Both S5 variants retain original binary integer halving, nominally a 50% reduction with floor rounding at base-unit precision. At a 60-second target interval, each long epoch lasts approximately four target years. The variants differ only in the boundary used as the phase anchor for post-activation calculations.

Both variants are deterministic, have no fixed terminal phase, and eventually reach zero through binary integer rounding.

4.7.1 S5/a - Long Epoch Anchored at Height 840,000

S5/a starts the first 2,100,000-block epoch at height 840,000. The subsidy therefore falls from 6.25 RIN at height 839,999 to 3.125 RIN at height 840,000. The next halving occurs at height 2,940,000.

S5/a keeps the immediate 50% miner-revenue shock at activation.

4.7.2 S5/b - Long Epoch Phase-Aligned to Height 630,000

For block heights at and above 840,000, the extended-epoch schedule is phase-aligned to height 630,000, the beginning of the existing 6.25 RIN subsidy period.

No block or validation rule below height 840,000 is changed. At this activation height, 210,000 blocks have elapsed since the phase anchor, so the subsidy remains 6.25000000 RIN through height 2,729,999. The first later halving is at height 2,730,000, approximately 3.60 target years after activation.

S5/b has no immediate miner-revenue shock but retains 50% reductions at later long-epoch boundaries.

4.8 S6 - Customized Halving Study Scenarios

S6 variants model the three Customized Halving scenarios in chapter 3 of Tokino's Rincoin v1.6.4 study. Tokino offers three possible scenarios I, II and III, which are included here as S6/a, S6/b, and S6/c ([Tokino 2026](#)).

Tokino's broader study combines deterministic block rewards with annual loss-rate assumptions, effective-circulating-supply modeling, a governance-controlled contingency that may switch Scenario II to Scenario III, and a view of subsidy design as subject to continuing governance control. This analysis retains those ideas as policy context but does not insert them into a fixed consensus function. The study does not specify a complete consensus authorization mechanism, deterministic trigger, or activation method for the contingency. Including it would prevent a fixed height-only comparison.

4.8.1 S6/a - Customized Halving Study Scenario I

S6/a follows the PDF's Scenario I reward path: 4 RIN from height 840,000, 2 RIN from 2,100,000, 1 RIN from 4,200,000, and a finite 0.5 RIN phase from 6,300,000 until the simulated 168,000,000 RIN ceiling is reached.

4.8.2 S6/b - Customized Halving Study Scenario II

S6/b follows the PDF’s Scenario II reward path: 4 RIN from height 840,000, 2 RIN from 2,100,000, 1 RIN from 4,200,000, and 0.6 RIN from height 6,300,000. For this bounded review interpretation, the final phase ends at the precomputed height needed to reach exactly 168,000,000 RIN maximum scheduled issuance: height 234,587,499 is the final non-zero block and height 234,587,500 is the first zero-subsidy block. This cutoff is provisional pending clarification from the study’s author. Underclaims are not reissued and do not extend the phase.

4.8.3 S6/c - Customized Halving Study Scenario III

S6/c follows the PDF’s Scenario III reward path: 4 RIN from height 840,000, 2 RIN from 2,100,000, 1 RIN from 4,200,000, an additional 1 RIN phase from 6,300,000, and a finite 0.6 RIN phase from 8,400,000 until the simulated 168,000,000 RIN ceiling is reached.

4.9 S7 - Aevust Core Customized Halving Tail-Emission Implementation

S7 models the inspected Aevust Core subsidy implementation snapshot as a separate source artifact. Its phase rewards correspond to Tokino Scenario II through height 6,300,000: 4 RIN from height 840,000, 2 RIN from 2,100,000, 1 RIN from 4,200,000, and 0.6 RIN from 6,300,000 ([Aevust 2026](#)).

The difference from S6/b is termination. S6/b is the review’s bounded height-only reconstruction of the published study direction. S7 models the available Core code path as inspected: no finite terminal height is visible in the subsidy function, so the 0.6 RIN phase behaves as tail emission unless another consensus rule elsewhere enforces a cap. The deterministic S6 study scenarios and the S7 code snapshot are separate artifacts and are not blended into a synthetic rule.

5 Deterministic Simulation Results

Generated artifacts use `data/scenario_config.json` as the machine-readable source of truth and `scripts/simulate_monetary_scenarios.py` as the generator. The year convention is 365 days and 525,600 target blocks per year. All consensus-deterministic quantities are integer base-unit outputs.

5.1 Scenario Summary

Sce- nario	Name	First post- 840,000 subsidy	Effective maximum scheduled issuance	Post-840,000 issuance	Zero height	Constant phase	Share of original 168M statement
S0	Unchanged binary halving	3.12500000	20999999.97690000	1312499.97690000	6930000	none	12.49999999
S1	1/20 reduction without a floor	5.93750000	44624993.01750000	24937493.01750000	73710000	none	26.56249584
S2/a	1/20 reduction with bounded 0.5 RIN phase	5.93750000	167999999.88020000	148312499.88020000	261919612	0.50000000	99.99999993
S2/b	Bounded 1 RIN phase comparator	5.93750000	167999999.31750000	148312499.31750000	135706704	1.00000000	99.99999959
S3	Perpetual 0.5 RIN tail-emission comparator	5.93750000	unbounded	unbounded	none	0.50000000	unbounded
S4	Idealized 113/114 cap-derived geometric schedule	6.19517543	167999782.97970000	148312282.97970000	384300000	none	99.99987082

Sce- nario	Name	First post- 840,000 subsidy	Effective maximum scheduled issuance	Post-840,000 issuance	Zero height	Constant phase	Share of original 168M statement
S5/a	Extended-epoch binary halving, 840,000 anchor	3.12500000	32812499.76900000	13124999.76900000	61740000	none	19.53124986
S5/b	Extended-epoch binary halving, phase-aligned to 630,000	6.25000000	44624999.76900000	24937499.76900000	63630000	none	26.56249986
S6/a	Customized Halving study scenario I	4.00000000	168000000.00000000	148312500.00000000	280245000	0.50000000	100.00000000
S6/b	Bounded Customized Halving study scenario II	4.00000000	168000000.00000000	148312500.00000000	234587500	0.60000000	100.00000000
S6/c	Customized Halving study scenario III	4.00000000	168000000.00000000	148312500.00000000	233187500	0.60000000	100.00000000
S7	Aevust Core Customized Halving tail-emission implementation	4.00000000	unbounded	unbounded	none	0.60000000	unbounded

The summary shows several distinct design families. S0 is the unchanged deployed baseline. S1 and both S5 variants are finite

schedules without a constant phase. S1 changes the reduction coefficient; S5/a and S5/b retain binary halving but use different phase anchors for post-activation 2,100,000-block epochs. S2 uses finite constant phases to approach the historical 168 million upper bound. S3 is a perpetual-tail comparator. S4 approaches the same upper bound through a shallow geometric coefficient. S6 contains Tokino’s finite Customized Halving study scenarios. S7 models the inspected Aevust Core implementation snapshot as an unbounded 0.6 RIN tail-emission variant.

The scenarios deliberately exercise different policy directions: deployed-code continuity, one-twentieth reduction, fixed-phase cap use, perpetual tail emission, cap-derived geometric decline, longer halving epochs, Tokino’s study scenarios and Aevust’s implementation snapshot. The deterministic tables show the cost of each direction in scheduled issuance, constant-phase duration, terminal height and near-term dilution.

5.2 Constant-Phase Results

Scenario	Starts	Ends	Duration, years	Subsidy	Issuance in phase	Share of final bounded issuance
S2/a	11130000	261919612	477.14918569	0.50000000	125394806.00000000	74.63976553
S2/b	8190000	135706704	242.61168950	1.00000000	127516704.00000000	75.90280031
S3	11130000	perpetual	perpetual	0.50000000	unbounded	unbounded
S6/a	6300000	280245000	521.20433790	0.50000000	136972500.00000000	81.53125000
S6/b	6300000	234587500	434.33694825	0.60000000	136972500.00000000	81.53125000
S6/c	8400000	233187500	427.67789193	0.60000000	134872500.00000000	80.28125000
S7	6300000	perpetual	perpetual	0.60000000	unbounded	unbounded

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The finite constant-phase scenarios are bounded, but the bound is reached only after very long durations. Duration of fixed phase of every assessed scenario substantially exceeds human life-span. It is also worth noting that the majority of the issuance is concentrated in the constant phase. Lowering the fixed phase from 1 RIN to 0.5 RIN lowers annual issuance but lengthens the phase when the goal is still to approach the 168 million ceiling. The scenarios have shorter or longer fixed phases depending on where the phase begins and which fixed reward is used.

5.3 Scarcity and Dilution Comparison

Scenario	Final or 500-year issuance	Supply multiple vs S0	Years to zero	Post-840,000 issuance after 1 year	After 10 years	After 25 years
S0	20999999.97690000	1.00000000	11.58675799	1066875.00000000	1312499.94384000	1312499.97690000
S1	44624993.01750000	2.12499967	138.64155251	2997273.75000000	18029979.34674000	23930272.32810000
S2/a	167999999.88020000	8.00000000	496.72681126	2997273.75000000	18029979.34674000	24342693.88020000
S2/b	167999999.31750000	7.99999998	256.59570776	2997273.75000000	18029979.34674000	26585795.31750000
S3	168860193.88020000 through 500y	unbounded	none	2997273.75000000	18029979.34674000	24342693.88020000
S4	167999782.97970000	7.99998967	729.56621005	3233345.00014800	29350378.35312000	62853444.64320000
S5/a	32812499.76900000	1.56249999	115.86757991	1642500.00000000	10668750.00000000	12946289.05980000
S5/b	44624999.76900000	2.12499999	119.46347032	3285000.00000000	20353125.00000000	24600585.93750000
S6/a	168000000.00000000	8.00000001	531.59246575	2102400.00000000	11136000.00000000	15180000.00000000
S6/b	168000000.00000000	8.00000001	444.72507610	2102400.00000000	11136000.00000000	15948000.00000000
S6/c	168000000.00000000	8.00000001	442.06145358	2102400.00000000	11136000.00000000	16788000.00000000
S7	185431500.00000000 through 500y	unbounded	none	2102400.00000000	11136000.00000000	15948000.00000000

For unbounded schedules, 500-year issuance is shown as a measurement horizon, not as a cap. The near-term columns are decision-relevant because the first 1, 10 and 25 years define the immediate dilution trade-off visible to holders and miners.

5.4 Subsidy Thresholds

The following table summarizes the subsidy thresholds for each scenario, i.e. the block heights (and estimated time from the activation height in years) at which the maximum permitted subsidy falls below 1 RIN, 0.5 RIN, and 0.1 RIN. The table also shows the final non-zero subsidy height for finite schedules and the final non-zero subsidy value for unbounded schedules.

Scenario	< 1 RIN	< 0.5 RIN	< 0.1 RIN	< 0.01 RIN	Zero subsidy
S0	1260000 (0.8 y)	1470000 (1.2 y)	1890000 (2.0 y)	2730000 (3.6 y)	6930000 (11.6 y)
S1	8190000 (14.0 y)	11130000 (19.6 y)	17640000 (32.0 y)	27090000 (49.9 y)	73710000 (138.6 y)
S2/a	8190000 (14.0 y)	261919612 (496.7 y)	261919612 (496.7 y)	261919612 (496.7 y)	261919612 (496.7 y)
S2/b	135706704 (256.6 y)	135706704 (256.6 y)	135706704 (256.6 y)	135706704 (256.6 y)	135706704 (256.6 y)
S3	8190000 (14.0 y)	never	never	never	none
S4	44310000 (82.7 y)	60900000 (114.3 y)	99330000 (187.4 y)	154140000 (291.7 y)	384300000 (729.6 y)
S5/a	5040000 (8.0 y)	7140000 (12.0 y)	11340000 (20.0 y)	19740000 (36.0 y)	61740000 (115.9 y)
S5/b	6930000 (11.6 y)	9030000 (15.6 y)	13230000 (23.6 y)	21630000 (39.6 y)	63630000 (119.5 y)
S6/a	6300000 (10.4 y)	280245000 (531.6 y)	280245000 (531.6 y)	280245000 (531.6 y)	280245000 (531.6 y)
S6/b	6300000 (10.4 y)	234587500 (444.7 y)	234587500 (444.7 y)	234587500 (444.7 y)	234587500 (444.7 y)
S6/c	8400000 (14.4 y)	233187500 (442.1 y)	233187500 (442.1 y)	233187500 (442.1 y)	233187500 (442.1 y)
S7	6300000 (10.4 y)	never	never	never	none

Threshold crossings expose both cliff risk and tail length. S0 falls below small nominal subsidies fastest. S1 has no fixed phase, but it still has a long declining geometric tail. S2 and S6 fixed-phase variants remain at or above their fixed subsidy until the terminal height. S3 and S7 remain non-zero indefinitely within the modelled horizon. S4 remains non-zero for the longest finite period. Both S5 variants retain the halving pattern; S5/b remains at each threshold longer than S5/a because its post-activation calculation is phase-aligned 210,000 blocks before activation.

5.5 Representative Horizon Outputs

Scenario	Horizon	Height	Subsidy	Next-year issuance	Annual issuance % of scheduled supply	Cumulative scheduled issuance	Remaining scheduled issuance
S0	activation	840000	3.12500000	1066875.00000000	5.41904762	19687500.00000000	1312499.97690000
S0	1 year	1365600	0.78125000	204726.56250000	0.98642605	20754375.00000000	245624.97690000
S0	2 years	1891200	0.09765625	33251.95180800	0.15865161	20959101.56250000	40898.41440000
S0	5 years	3468000	0.00076293	198.32205600	0.00094440	20999761.95534000	238.02156000
S0	10 years	6096000	0.00000009	0.02899200	0.00000014	20999999.94384000	0.03306000
S0	25 years	13980000	0.00000000	0.00000000	0.00000000	20999999.97690000	0.00000000
S0	100 years	53400000	0.00000000	0.00000000	0.00000000	20999999.97690000	0.00000000
S0	terminal height	6930000	0.00000000	0.00000000	0.00000000	20999999.97690000	0.00000000
S1	activation	840000	5.93750000	2997273.75000000	15.22424762	19687500.00000000	24937493.01750000
S1	1 year	1365600	5.35859375	2649577.30776000	11.67998119	22684773.75000000	21940219.26750000
S1	2 years	1891200	4.59432430	2318695.74579600	9.15237868	25334351.05776000	19290641.95974000
S1	5 years	3468000	3.20838798	1585301.55716400	5.03329838	31496276.17164000	13128716.84586000
S1	10 years	6096000	1.64700054	830448.96382800	2.20176157	37717479.34674000	6907513.67076000
S1	25 years	13980000	0.24686925	121558.42755600	0.27869013	43617772.32810000	1007220.68940000
S1	100 years	53400000	0.00001592	7.94121600	0.00001780	44624929.03650000	63.98100000
S1	terminal height	73710000	0.00000000	0.00000000	0.00000000	44624993.01750000	0.00000000
S2/a	activation	840000	5.93750000	2997273.75000000	15.22424762	19687500.00000000	148312499.88020000
S2/a	1 year	1365600	5.35859375	2649577.30776000	11.67998119	22684773.75000000	145315226.13020000
S2/a	2 years	1891200	4.59432430	2318695.74579600	9.15237868	25334351.05776000	142665648.82244000

Sce- nario	Horizon	Height	Subsidy	Next-year issuance	Annual issuance % of scheduled supply	Cumulative scheduled issuance	Remaining scheduled issuance
S2/a	5 years	3468000	3.20838798	1585301.55716400	5.03329838	31496276.17164000	136503723.70856000
S2/a	10 years	6096000	1.64700054	830448.96382800	2.20176157	37717479.34674000	130282520.53346000
S2/a	25 years	13980000	0.50000000	262800.00000000	0.59686315	44030193.88020000	123969806.00000000
S2/a	100 years	53400000	0.50000000	262800.00000000	0.41229871	63740193.88020000	104259806.00000000
S2/a	terminal height	261919612	0.00000000	0.00000000	0.00000000	167999999.88020000	0.00000000
S2/b	activation	840000	5.93750000	2997273.75000000	15.22424762	19687500.00000000	148312499.31750000
S2/b	1 year	1365600	5.35859375	2649577.30776000	11.67998119	22684773.75000000	145315225.56750000
S2/b	2 years	1891200	4.59432430	2318695.74579600	9.15237868	25334351.05776000	142665648.25974000
S2/b	5 years	3468000	3.20838798	1585301.55716400	5.03329838	31496276.17164000	136503723.14586000
S2/b	10 years	6096000	1.64700054	830448.96382800	2.20176157	37717479.34674000	130282519.97076000
S2/b	25 years	13980000	1.00000000	525600.00000000	1.13586032	46273295.31750000	121726704.00000000
S2/b	100 years	53400000	1.00000000	525600.00000000	0.61335020	85693295.31750000	82306704.00000000
S2/b	terminal height	135706704	0.00000000	0.00000000	0.00000000	167999999.31750000	0.00000000
S3	activation	840000	5.93750000	2997273.75000000	15.22424762	19687500.00000000	
S3	1 year	1365600	5.35859375	2649577.30776000	11.67998119	22684773.75000000	
S3	2 years	1891200	4.59432430	2318695.74579600	9.15237868	25334351.05776000	
S3	5 years	3468000	3.20838798	1585301.55716400	5.03329838	31496276.17164000	
S3	10 years	6096000	1.64700054	830448.96382800	2.20176157	37717479.34674000	
S3	25 years	13980000	0.50000000	262800.00000000	0.59686315	44030193.88020000	
S3	100 years	53400000	0.50000000	262800.00000000	0.41229871	63740193.88020000	
S4	activation	840000	6.19517543	3233345.00014800	16.42333968	19687500.00000000	148312282.97970000

Scenario	Horizon	Height	Subsidy	Next-year issuance	Annual issuance % of scheduled supply	Cumulative scheduled issuance	Remaining scheduled issuance
S4	1 year	1365600	6.08696483	3165578.04270000	13.81091335	22920845.00014800	145078937.97955200
S4	2 years	1891200	5.92818254	3093873.57891600	11.86009126	26086423.04284800	141913359.93685200
S4	5 years	3468000	5.57361171	2898255.42720000	8.24077756	35169684.01368000	132830098.96602000
S4	10 years	6096000	4.97042362	2593609.72623600	5.28899254	49037878.35312000	118961904.62658000
S4	25 years	13980000	3.58770144	1864467.11114400	2.25883908	82540944.64320000	85458838.33650000
S4	100 years	53400000	0.68462278	356596.29661200	0.23514245	151651180.10040000	16348602.87930000
S4	terminal height	384300000	0.00000000	0.00000000	0.00000000	167999782.97970000	0.00000000
S5/a	activation	840000	3.12500000	1642500.00000000	8.34285714	19687500.00000000	13124999.76900000
S5/a	1 year	1365600	3.12500000	1642500.00000000	7.70042194	21330000.00000000	11482499.76900000
S5/a	2 years	1891200	3.12500000	1642500.00000000	7.14985309	22972500.00000000	9839999.76900000
S5/a	5 years	3468000	1.56250000	821250.00000000	3.03324100	27075000.00000000	5737499.76900000
S5/a	10 years	6096000	0.78125000	410625.00000000	1.35268684	30356250.00000000	2456249.76900000
S5/a	25 years	13980000	0.04882812	25664.05987200	0.07864260	32633789.05980000	178710.70920000
S5/a	100 years	53400000	0.00000009	0.04730400	0.00000014	32812499.43840000	0.33060000
S5/a	terminal height	61740000	0.00000000	0.00000000	0.00000000	32812499.76900000	0.00000000
S5/b	activation	840000	6.25000000	3285000.00000000	16.68571429	19687500.00000000	24937499.76900000
S5/b	1 year	1365600	6.25000000	3285000.00000000	14.29970617	22972500.00000000	21652499.76900000
S5/b	2 years	1891200	6.25000000	3285000.00000000	12.51071123	26257500.00000000	18367499.76900000
S5/b	5 years	3468000	3.12500000	1642500.00000000	4.85856905	33806250.00000000	10818749.76900000
S5/b	10 years	6096000	1.56250000	821250.00000000	2.05104191	40040625.00000000	4584374.76900000
S5/b	25 years	13980000	0.09765625	51328.12500000	0.11589601	44288085.93750000	336913.83150000

Scenario	Horizon	Height	Subsidy	Next-year issuance	Annual issuance % of scheduled supply	Cumulative scheduled issuance	Remaining scheduled issuance
S5/b	100 years	53400000	0.00000018	0.09460800	0.00000021	44624999.10360000	0.66540000
S5/b	terminal height	63630000	0.00000000	0.00000000	0.00000000	44624999.76900000	0.00000000
S6/a	activation	840000	4.00000000	2102400.00000000	10.67885714	19687500.00000000	148312500.00000000
S6/a	1 year	1365600	4.00000000	2102400.00000000	9.64850688	21789900.00000000	146210100.00000000
S6/a	2 years	1891200	4.00000000	1468800.00000000	6.14758730	23892300.00000000	144107700.00000000
S6/a	5 years	3468000	2.00000000	1051200.00000000	3.82762576	27463500.00000000	140536500.00000000
S6/a	10 years	6096000	1.00000000	364800.00000000	1.18351258	30823500.00000000	137176500.00000000
S6/a	25 years	13980000	0.50000000	262800.00000000	0.75371048	34867500.00000000	133132500.00000000
S6/a	100 years	53400000	0.50000000	262800.00000000	0.48151711	54577500.00000000	113422500.00000000
S6/a	terminal height	280245000	0.00000000	0.00000000	0.00000000	168000000.00000000	0.00000000
S6/b	activation	840000	4.00000000	2102400.00000000	10.67885714	19687500.00000000	148312500.00000000
S6/b	1 year	1365600	4.00000000	2102400.00000000	9.64850688	21789900.00000000	146210100.00000000
S6/b	2 years	1891200	4.00000000	1468800.00000000	6.14758730	23892300.00000000	144107700.00000000
S6/b	5 years	3468000	2.00000000	1051200.00000000	3.82762576	27463500.00000000	140536500.00000000
S6/b	10 years	6096000	1.00000000	396960.00000000	1.28784856	30823500.00000000	137176500.00000000
S6/b	25 years	13980000	0.60000000	315360.00000000	0.88496022	35635500.00000000	132364500.00000000
S6/b	100 years	53400000	0.60000000	315360.00000000	0.53191651	59287500.00000000	108712500.00000000
S6/b	terminal height	234587500	0.00000000	0.00000000	0.00000000	168000000.00000000	0.00000000
S6/c	activation	840000	4.00000000	2102400.00000000	10.67885714	19687500.00000000	148312500.00000000
S6/c	1 year	1365600	4.00000000	2102400.00000000	9.64850688	21789900.00000000	146210100.00000000

Scenario	Horizon	Height	Subsidy	Next-year issuance	Annual issuance % of scheduled supply	Cumulative scheduled issuance	Remaining scheduled issuance
S6/c	2 years	1891200	4.00000000	1468800.00000000	6.14758730	23892300.00000000	144107700.00000000
S6/c	5 years	3468000	2.00000000	1051200.00000000	3.82762576	27463500.00000000	140536500.00000000
S6/c	10 years	6096000	1.00000000	525600.00000000	1.70519247	30823500.00000000	137176500.00000000
S6/c	25 years	13980000	0.60000000	315360.00000000	0.86458033	36475500.00000000	131524500.00000000
S6/c	100 years	53400000	0.60000000	315360.00000000	0.52448547	60127500.00000000	107872500.00000000
S6/c	terminal height	233187500	0.00000000	0.00000000	0.00000000	168000000.00000000	0.00000000
S7	activation	840000	4.00000000	2102400.00000000	10.67885714	19687500.00000000	
S7	1 year	1365600	4.00000000	2102400.00000000	9.64850688	21789900.00000000	
S7	2 years	1891200	4.00000000	1468800.00000000	6.14758730	23892300.00000000	
S7	5 years	3468000	2.00000000	1051200.00000000	3.82762576	27463500.00000000	
S7	10 years	6096000	1.00000000	396960.00000000	1.28784856	30823500.00000000	
S7	25 years	13980000	0.60000000	315360.00000000	0.88496022	35635500.00000000	
S7	100 years	53400000	0.60000000	315360.00000000	0.53191651	59287500.00000000	

The complete horizon table for all scenarios is generated at `data/horizon_table.csv`.

6 Interpretation, Scoring, and Review-Set Selection

Note: because S1, S2 and S3 have exactly the same progress in the first ~14 years, their graphs overlap for those three scenarios in the given period and is visible as S3. The same is true for the S6 and S7 in first ~10 years (visible as S7).

6.1 Monetary Figures

The 500-year figures are structural comparisons. Unbounded scenarios are truncated at the 500-year plot horizon.

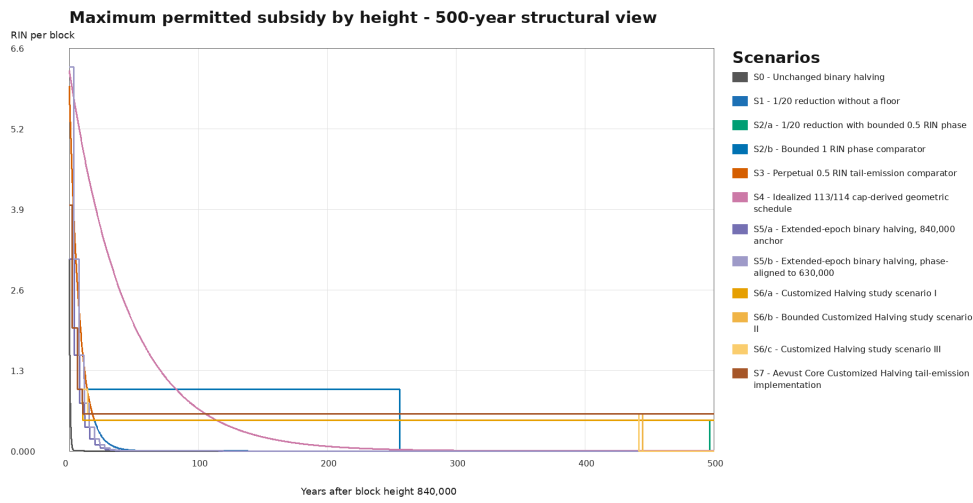


Figure 1: Maximum permitted subsidy, 500-year structural view

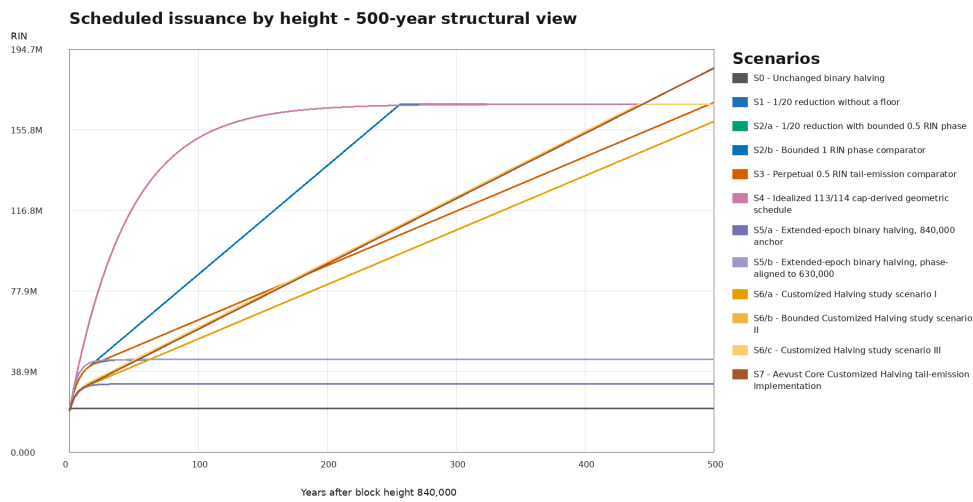


Figure 2: Scheduled issuance, 500-year structural view

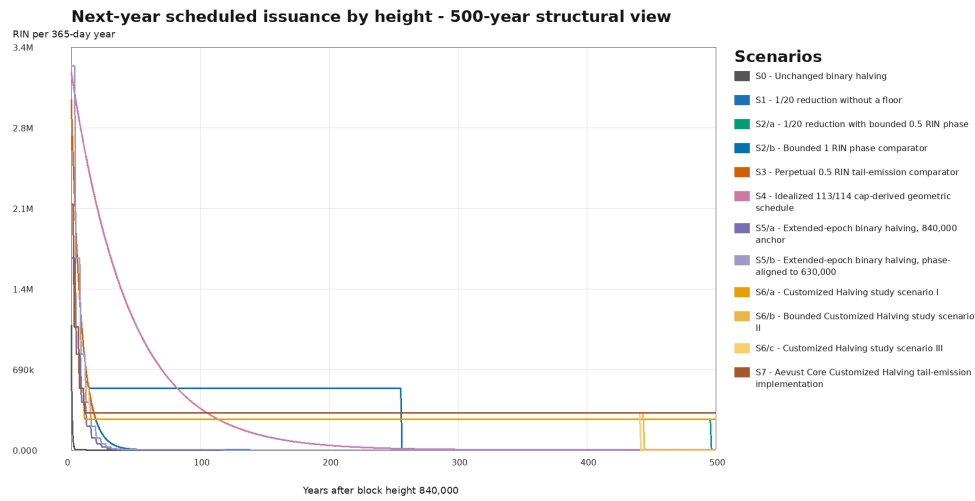


Figure 3: Annual scheduled issuance, 500-year structural view

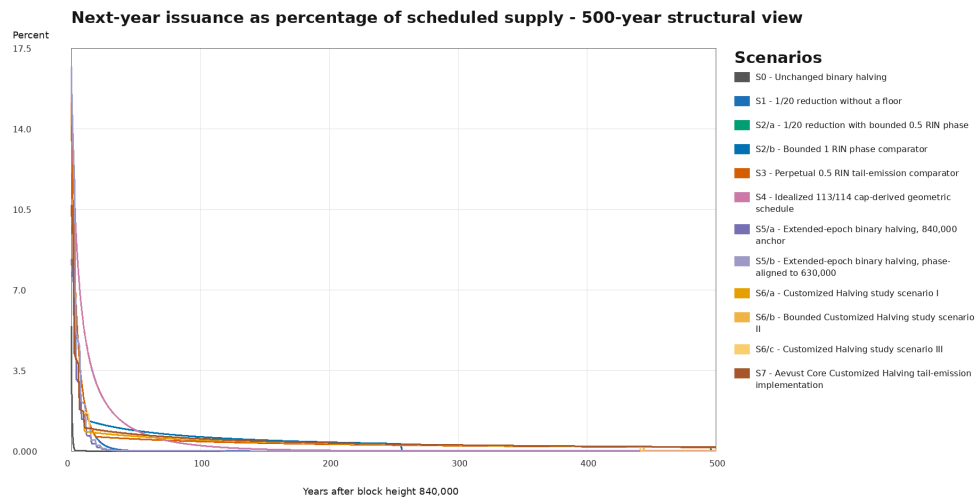


Figure 4: Annual scheduled issuance as percentage of scheduled supply, 500-year structural view

Near-term figures show differences that are visually compressed in the 500-year view.

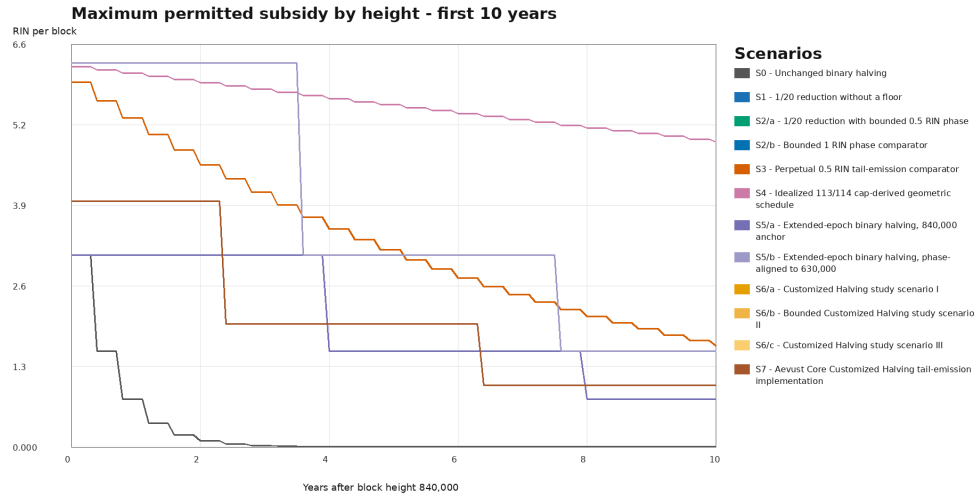


Figure 5: Maximum permitted subsidy, first 10 years

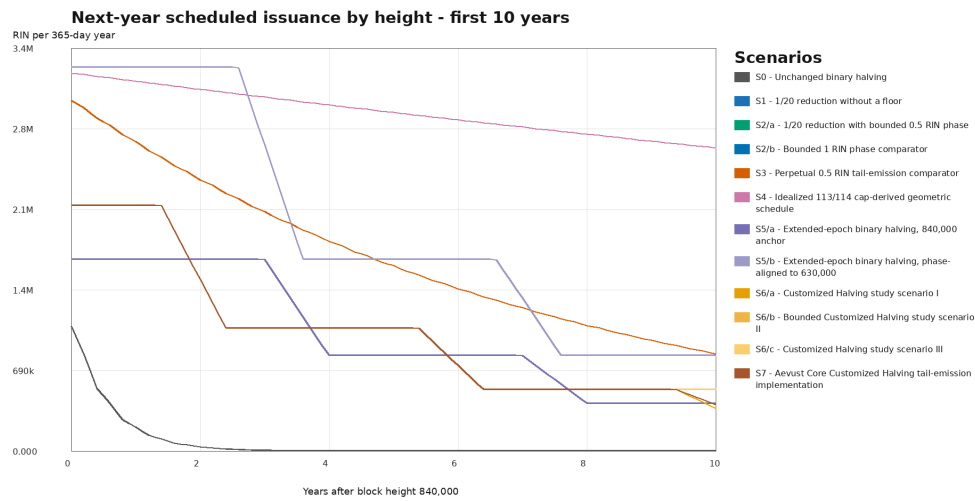


Figure 6: Annual scheduled issuance, first 10 years

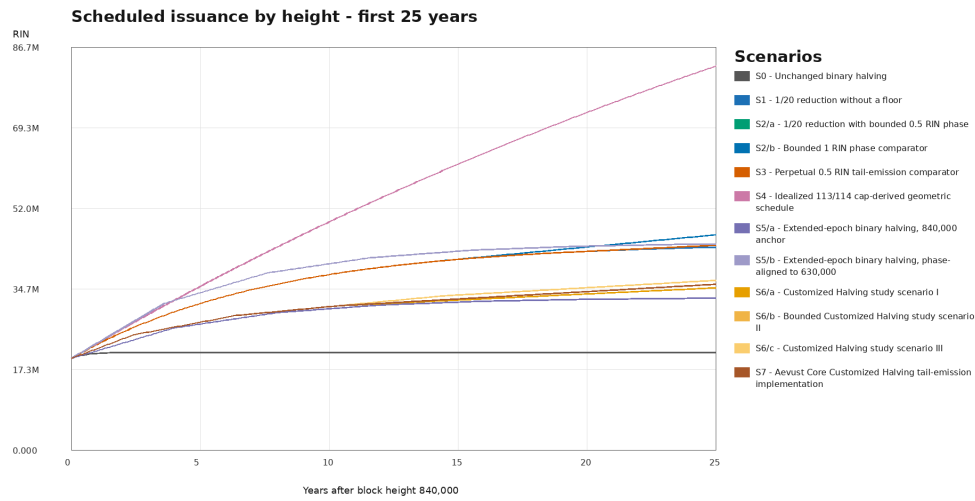


Figure 7: Scheduled issuance, first 25 years

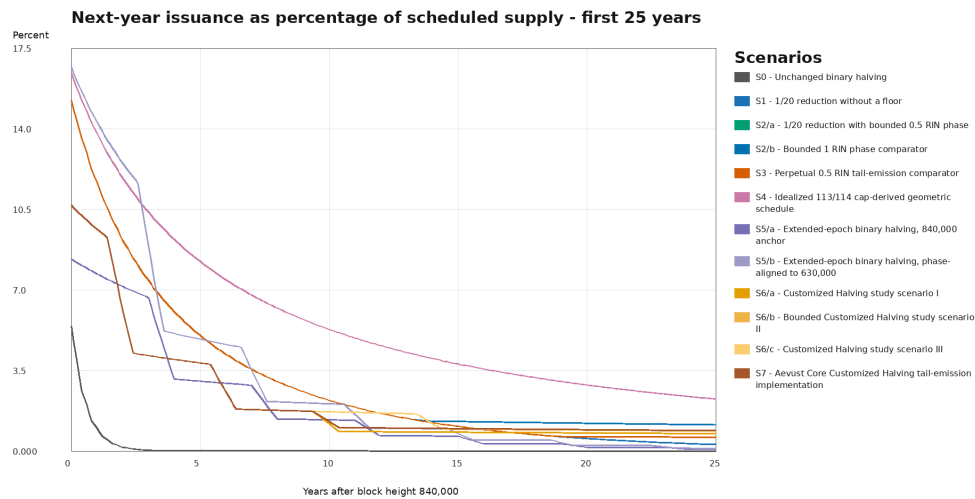


Figure 8: Annual scheduled issuance as percentage of scheduled supply, first 25 years

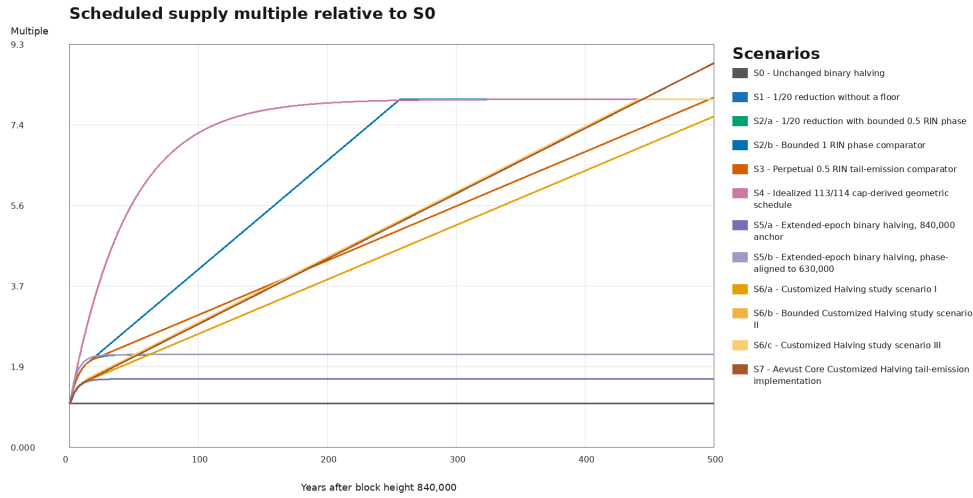


Figure 9: Scheduled supply multiple relative to S0, 500-year structural view

6.2 Comparative Evaluation

The deterministic results reveal four broad outcomes.

- **S0** minimizes issuance and implementation change, but it leaves the five-month binary-halving cadence unchanged. It is therefore the baseline against which a correction must be justified, not automatically the best corrective policy.
- **S1 and the two S5 variants** are finite schedules without a constant phase. S1 replaces every 50% cliff with a 5% reduction and preserves subsidy above 1 RIN for about 14 years. S5 variants keep the halving reduction but prolong epochs ten times. S5/a has the lowest ten-year and final issuance of the three, but cuts subsidy by 50% at activation. S5/b has no immediate miner-revenue shock and remains above 1 RIN for about 11.59 years, but schedules more supply in the first ten years than S1 and retains delayed 50% cuts.
- **S4** provides the smoothest declining path and a long nominal runway, but it schedules almost the full historical 168M amount, creates about 149% additional issuance relative to activation supply within ten years, and continues for roughly 730 years. The 113/114 coefficient is mathematically auditable but less intuitive to explain than one twentieth or a conventional halving interval.
- **S2 and S6** are finite, but their constant phases create most of their final issuance and last for centuries.
- **S3 and S7** preserve a permanent nominal subsidy at the cost of unbounded Layer-1 issuance.

The original whitepaper's documented 168,000,000 RIN maximum-supply statement is not scored as a separate virtue. Treating it as a strict target, a maximum

ceiling, or a historical parameter inconsistency is a policy interpretation that should be stated openly and discussed before it is used to justify any particular scenario.

Scenario	Main strength	Main weakness or trade-off
S0	Lowest issuance, no hard fork, simplest continuation	Does not correct the rapid 50% reduction cadence; subsidy falls below 1 RIN in less than one year
S1	Gradual 5% epoch reductions; finite 44.625M maximum; no fixed phase	Frequent monetary changes, weak major-halving salience, very low to none subsidy over a decades-long horizon
S2/a	Smooth early decline and finite issuance	About 75% of final issuance occurs in a roughly 477-year 0.5 RIN phase
S2/b	Long nominal runway and finite issuance	About 76% of final issuance occurs in a roughly 243-year 1 RIN phase
S3	Simple permanent nominal subsidy	Unbounded supply
S4	Very smooth decline and long nominal runway	Near-168M issuance, high near-term dilution, 113/114 explanation burden and roughly 730 years to zero
S5/a	Simple rule, 32.8125M maximum, lower ten-year scheduled supply expansion than S1	Immediate 50% activation cut and subsidy below 1 RIN after about 7.99 years, very low to none subsidy over a decades-long horizon
S5/b	No immediate miner-revenue shock; familiar four-year rhythm; 44.625M maximum similar to S1	About 103% ten-year scheduled expansion and first later 50% cut after about 3.60 years, very low to none subsidy over a decades-long horizon
S6/a	Finite implementation of the published 0.5 RIN study path	Multiple phases and a roughly 521-year cap-filling constant phase
S6/b	Finite implementation of the published 0.6 RIN study path	Multiple phases and a roughly 434-year cap-filling constant phase
S6/c	Longer 1 RIN phase before the final 0.6 RIN phase	Multiple phases and a roughly 428-year cap-filling constant phase
S7	Long-lived nominal subsidy in inspected Core code	Unbounded issuance and several manually selected phase boundaries

6.3 Eligibility Checks

The scenarios were evaluated against these checks to determine their eligibility for scoring.

Scenario	Deterministic rule	Corrective relevance	Bounded issuance	Scoring status
S0	✓ Satisfied	✗ Baseline outside corrective candidate set	✓ Satisfied	✗ Baseline outside corrective candidate set
S1	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S2/a	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S2/b	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S3	✓ Satisfied	✓ Satisfied	✗ Outside bounded-supply candidate set	✗ Outside bounded-supply candidate set
S4	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S5/a	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S5/b	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S6/a	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S6/b	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S6/c	✓ Satisfied	✓ Satisfied	✓ Satisfied	✓ Eligible
S7	✓ Satisfied	✓ Satisfied	✗ Outside bounded-supply candidate set	✗ Outside bounded-supply candidate set

The scenario S0 was opted out of the candidate scoring set because it is the unchanged baseline. Scenarios S3 and S7 were opted out because they are unbounded and therefore do not meet the fixed-supply requirement.

6.4 Evaluation by Criteria

The scored table contains only scenarios that pass all three eligibility checks. S0 remains the deployed baseline. S3 and S7 remain security-economics comparators. None of those three is included in the candidate balance total.

The symbols mean: ✓ = 4 points, strong fit; ♦ = 2-3 points, mixed or good fit with a trade-off; ✗ = 0-1 point, weak fit or failure.

Criterion	S1	S2/a	S2/b	S4	S5/a	S5/b	S6/a	S6/b	S6/c
Reward-path correction	4 ✓	4 ✓	4 ✓	4 ✓	3 ♦	3 ♦	2 ♦	2 ♦	2 ♦
Monetary discipline	4 ✓	1 ✗	1 ✗	2 ♦	4 ✓	4 ✓	1 ✗	1 ✗	1 ✗
Ten-year scheduled supply expansion	2 ♦	2 ♦	2 ♦	1 ✗	3 ♦	1 ✗	3 ♦	3 ♦	3 ♦
Nominal subsidy runway	3 ♦	3 ♦	4 ✓	4 ✓	2 ♦	3 ♦	3 ♦	3 ♦	3 ♦
Rule clarity and auditability	4 ✓	2 ♦	2 ♦	3 ♦	4 ✓	4 ✓	1 ✗	1 ✗	1 ✗
Balance indicator, unweighted	17/20	12/20	13/20	14/20	16/20	15/20	10/20	10/20	10/20

6.4.1 Why the Scores Differ

- **S1 scores 17/20** because 5% recurring reductions meet the highest reward-path-correction band, while its finite supply, nominal runway, and one-formula rule score well under the stated thresholds.
- **S5/a scores 16/20** because it is simple, scarce and schedules less near-term supply than S1. It remains one point lower because it preserves 50% cliffs and reaches sub-1-RIN subsidy sooner.
- **S5/b scores 15/20** because it is equally simple, has no immediate miner-revenue shock, and has a longer nominal runway than S5/a. Its ten-year scheduled supply expansion is just over 100% of pre-activation scheduled supply, reducing that criterion from two points to one under the current threshold. Immediate shock and milestone value are evaluated outside the monetary score rather than counted as hidden criteria.
- **S4 scores 14/20** because it is smooth and long-lived, but near-term scheduled supply expansion is high, final issuance is near 168M, and the 113/114 rule is less immediately legible.
- **S2/a and S2/b score 12/20 and 13/20** because their early path is smooth and bounded, but the cap-filling constant phase dominates final issuance and complicates the rule.
- **S6 variants score 10/20** because they provide a long nominal runway and moderate ten-year scheduled supply expansion, but use several phase boundaries and derive most issuance from centuries-long constant phases.

The point totals are compact policy indicators, not an objective ranking of all relevant considerations. They omit the milestone and social-coordination dimension, and small changes in a category boundary can change a total without changing any monetary result:

S1's ten-year scheduled supply expansion is 91.58% of pre-activation scheduled supply; S5/b's is 103.38%. The current two-point interval ends at 100.00%, so S1 receives 2 points and S5/b receives 1. Moving only that boundary to 105.00% would increase S5/b by one point without changing any monetary result. Numerical totals are compact policy indicators, not objective proof of superiority, and they do not include milestone or social-coordination considerations.

7 Evaluation Conclusions and Review-Set Selection

For the principal public-review set, scenarios **S1**, **S5/b** and **S6/b** have been selected for two different reasons. S1 and S5/b form a controlled near-equal-total-issuance pair despite their different raw scores. S6/b is the bounded representative of the published Customized Halving direction and related implementation history; it is not included because its score equals the leading schedules. S5/a remains the second-highest-scoring secondary comparator. S0 remains the deployed baseline. Numerical scoring is one selection input, not the only mechanism, and none of these labels adopts or prefers a candidate.

The unscored scenarios still matter. S0 measures the cost of taking no corrective action. S3 and S7 show the security-budget appeal and monetary cost of unbounded issuance. They should be included in deterministic and security-economics comparisons, but not allowed to recover eligibility through points after failing a declared gate.

7.1 Monetary Milestones, Market Psychology and Social Coordination

Discrete halving events may have significance beyond their arithmetic effect. They can create predictable public milestones, concentrate community and media attention, renew discussion of monetary scarcity, coordinate software releases and service readiness, and provide a familiar long-term narrative. Bitcoin halvings are widely associated in public discussion with scarcity narratives and later bull-market cycles, but temporal association and self-reinforcing expectations do not establish that halvings cause price increases. A recent synthetic-control preprint reports a positive estimate for the 2024 Bitcoin halving but no statistically significant and robust estimate for the 2020 halving ([Virtanen 2025](#)).

The same salience can create speculative volatility, excessive expectations, disappointment when price does not respond, abrupt miner-revenue shocks, and concentrated operational risk at a known boundary. These are forward-looking social and market hypotheses, not consensus-deterministic simulation outputs. They are not assigned a numerical score.

Under S1, every epoch boundary is technically predictable, but a 5% adjustment is unlikely to become a major public event by itself. S1 offers continuous monetary adjustment rather than rare dramatic milestones. Releases, annual reports, and anniversaries could still provide community milestones, but they would not carry the same consensus-level scarcity signal.

Under S5/b, the approximately four-year cadence is simple and familiar. Each halving could function as a major community and market milestone, support attention and anticipation, and provide a recognizable scarcity narrative. The same visibility may magnify volatility, disappointment, or miner exit when the 50% boundary arrives.

Dimension	S1	S5/b
Deterministic predictability	Strong	Strong
Subsidy-path smoothness	High	Low
Frequency of monetary boundaries	High	Low
Individual event salience	Low	High
Potential concentration of event-driven volatility	Low	High
Miner-revenue discontinuity	Low per boundary	High at later halvings
Time before first post-activation reduction	Approximately 0.40 target years	Approximately 3.60 target years

7.2 S1 and S5/b as an Equal-Total-Issuance Comparison

S1 and S5/b allocate nearly the same total post-840,000 issuance budget through different time paths. The ideal real-number derivation and integer-consensus results are generated from the scenario configuration:

For S1, the ideal real-number post-840,000 total is:

$$210000 * 5.9375 / (1 - 0.95) = 24,937,500 \text{ RIN}$$

For S5/b, the ideal real-number post-840,000 total is:

$$1890000 * 6.25 + 2100000 * 3.125 / (1 - 0.5) = 24,937,500 \text{ RIN}$$

Integer consensus arithmetic produces:

- S1 maximum scheduled issuance: 44,624,993.01750000 RIN;
- S5/b maximum scheduled issuance: 44,624,999.76900000 RIN;
- S5/b minus S1: 6.75150000 RIN.

The ideal post-activation totals are equal. The small integer-result difference comes from their different rounding paths. The policy choice is therefore primarily about issuance timing and shape rather than final amount: S1 uses recurring 19/20 reductions, nominally 5%, while S5/b front-loads more issuance and then applies four-year binary integer halvings, nominally 50%.

The 6.75150000 RIN integer difference is negligible relative to either schedule's total. The decision is therefore primarily about when issuance occurs and how miners, markets, and services encounter reductions. S1 distributes the remaining issuance through recurring 5% reductions. S5/b initially front-loads more issuance and then distributes the remainder through approximately four-year 50% halvings.

Metric	S1	S5/b
Subsidy at activation	5.93750000 RIN	6.25000000 RIN
Immediate miner-revenue shock	5%	0%
Consensus divergence height	840,000	840,000
First later reduction	height 1,050,000	height 2,730,000
First later reduction amount	5%	50%
Nominal ordinary reduction	5%	50%
Cadence	210,000 blocks	2,100,000 blocks
Mathematical half-life	13.51340733 epochs	1.00000000 epochs
Mathematical half-life	~5.4 target years	~4.0 target years
Post-activation issuance after 1 year	2997273.75000000 RIN	3285000.00000000 RIN
Post-activation issuance after 5 years	11808776.17164000 RIN	14118750.00000000 RIN
Post-activation issuance after 10 years	18029979.34674000 RIN	20353125.00000000 RIN
Post-activation issuance after 25 years	23930272.32810000 RIN	24600585.93750000 RIN
Maximum scheduled issuance	44624993.01750000 RIN	44624999.76900000 RIN
First zero-subsidy height	73,710,000	63,630,000

7.3 S6/b as the Bounded Customized Halving Representative

S6/b enters the principal review set for a different reason from the S1 + S5/b controlled pair. It is the deterministic bounded representative of the published Customized Halving Scenario II direction and related implementation history. Its inclusion is representational, not an inference that its 10/20 monetary score equals the higher-scoring schedules. S5/a remains the second-highest-scoring secondary comparator.

The review interpretation retains the study's 4, 2, 1 and 0.6 RIN phase amounts but fixes the consensus comparison as a height-only bounded schedule. The 0.6 RIN phase ends after 228,287,500 blocks: height 234,587,499 is the final non-zero-subsidy block and height 234,587,500 is the first zero-subsidy block. Maximum scheduled issuance is exactly 168,000,000 RIN. Underclaims are not reissued and do not change that height.

This later reconstruction makes the review candidate implementable but does not retroactively improve the source proposal's rule-clarity score or supply a source derivation for the phase values, transition heights, governance trigger or terminal behavior. The separate S7 code snapshot remains unbounded and must not be blended with S6/b.

8 Security Economics Analysis

8.1 Security-Economics Scope and Definitions

Deterministic subsidy schedules describe maximum issuance, not realised network security. A subsidy becomes miner revenue only in combination with transaction fees, market value and sufficient liquidity. Hashrate then depends on mining efficiency, operating cost, available hardware, miner expectations and alternative uses of that hardware. A higher nominal subsidy may support more mining revenue under otherwise equal conditions, but it does not guarantee hashrate, decentralisation or resistance to reorganisation.

For a period containing (B) blocks, maximum subsidy (R) RIN per block, total transaction fees (F) RIN per block and price (P), gross miner revenue is:

$$V_{\text{gross}} = B(R + F)P.$$

Fee-market analysis uses **total fees per block**. Average fee per transaction is reported separately and is not substituted for block fee revenue. The honest-chain security budget is expected revenue from extending the accepted chain. Rental expense, electricity expense, hashrate and attack cost are related but distinct: none alone establishes the probability, duration or profitability of a reorganisation.

Results in this chapter are labelled as:

1. **consensus-deterministic**;
2. **directly observed from a frozen public source**;
3. **derived from observed data**;
4. **operator-supplied evidence**;
5. **normalised sensitivity**; or
6. **speculative stress case**.

Operator-supplied evidence is retained when it changes a relevant sensitivity case, but it is never presented as independently verified. The core chain snapshot is frozen at block height 674,074. A separately frozen market snapshot collected on 2026-07-27 contains exact halving blocks, CoinPaprika's maximum free daily-history window, 60 rate-limited NestEx tradebook pages, and supplemental node data. Document regeneration reads only these files and performs no live API calls ([Rincoin Community Forge 2026a](#), [2026b](#)).

8.2 Empirical Current-Network Snapshot

Metric	Value	Unit	Evidence class
Local-node block height	674074	blocks	directly observed
Maximum current block subsidy	6.25000000	RIN/block	consensus-deterministic and directly observed
Current difficulty	0.333779833148	difficulty	directly observed
Current network hashrate estimate	19.70723840	MH/s	directly observed node estimate
NestEx RIN price	0.000656840000	USDT/RIN	directly observed venue quote
Scheduled daily issuance	9000.00000000	RIN/day	derived consensus-deterministic
Observed non-coinbase transactions (90 d)	6362	transactions	derived from observed data
Mean total fees per block (90 d)	0.00010003	RIN/block	derived from observed data
Fee share of miner revenue (90 d)	0.00097646	%	derived from observed data
Explorer-indexed orphaned blocks (90 d)	29	blocks	directly observed explorer index
Visible RinHash rental capacity	10.346300	MH/s	directly observed asks
Visible rental capacity / current node hashrate	52.50	%	derived from observed data
NestEx bid depth within 5%	18440.91901400	RIN	derived from observed order book

The fully synchronized local node and explorer agreed at block height 674,074. The maximum permitted subsidy was 6.25 RIN, point-in-time difficulty was 0.333779833148, and the node's current network-hashrate estimate was 19.70723840 MH/s. The 60-second target implies 1,440 target blocks and a maximum scheduled 9,000 RIN per day at that height.

The frozen prices must not be merged as though they represented one market. The explorer reported 0.000537652849 USD/RIN, while NestEx reported 0.000656840000 USDT/RIN. At those marginal prices, scheduled daily subsidy had a notional value of about 4.84 USD or 5.91 USDT respectively; neither figure represents realisable sale proceeds.

The 90-day mean total fee was 0.00010003 RIN per block, the median was zero, and fees were approximately 0.000976% of aggregate subsidy-plus-fee revenue.

Visible Mining Rig Rentals asks advertised 10.3463 MH/s, equivalent to 52.5% of the node’s current honest-network estimate before combining the two.

The source URLs, retrieval timestamps, response hashes, and transformations for these observations are frozen in the empirical evidence manifest ([Rincoin Community Forge 2026a](#)).

8.3 Historical Network and Fee Baseline

Metric	30 days	90 days	365 days	Unit
Blocks observed	40,506	122,084	496,274	blocks
Actual blocks/day	1350.19	1356.48	1359.65	blocks/day
Rolling hashrate, median	16.316	16.386	26.976	MH/s
Rolling hashrate, p10-p90	14.275-18.990	13.213-20.575	15.565-54.176	MH/s
Difficulty, median	0.229016	0.229838	0.400615	difficulty
Block interval, median	39.0	40.0	41.0	seconds
Block interval, p10-p90	6.0-153.0	7.0-152.0	6.0-149.0	seconds
Non-coinbase transactions	1,862	6,362	68,946	transactions
Blocks with no non-coinbase transaction	95.71	95.14	89.73	%
Total fee/block, mean	0.00006040	0.00010003	0.00019188	RIN/block
Total fee/block, median	0.00000000	0.00000000	0.00000000	RIN/block
Fee/non-coinbase transaction, mean	0.00131397	0.00191945	0.00138117	RIN/transaction
Fee/non-coinbase transaction, median	0.00080000	0.00099999	0.00028372	RIN/transaction
Aggregate fee share of miner revenue	0.000966	0.000976	0.000976	%
Indexed orphaned blocks	14	29	29	blocks
Indexed orphan incidence	0.034551	0.023748	0.005843	%
Indexed orphan reorg depth, maximum	1	1	1	blocks

The hashrate series is derived from chainwork accumulated across rolling 120-block windows. This avoids interpreting each short timestamp interval as a stable network-hashrate measurement. Over 90 days, median rolling hashrate was 16.386 MH/s, with p10 and p90 values of 13.213 and 20.575 MH/s. The chain produced about 1,356.5 blocks per day over that window, below the 1,440-block target, and the median observed header-time interval was 40 seconds. Header timestamps can move backwards; the raw 365-day minimum interval is therefore negative and is retained rather than silently clipped.

Transaction use was sparse. The 90-day window contained 6,362 non-coinbase transactions, and 95.14% of blocks contained none. Exact transaction fees were

reconstructed as previous-output value minus output value for 68,946 transactions in the 365-day dataset. Every reconstructed block total matched `getblockstats totalfee`. This permits an exact transaction-fee median while retaining total fees per block as the miner-revenue measure.

The explorer's new orphan index returned 29 records in total, including 14 in the last 30 days; all had reorg depth 1. Indexed orphan incidence was approximately 0.034551% over 30 days and 0.023749% over 90 days. The endpoint exhausted below its 100-record page limit, but database retention and pre-index completeness were not independently reconstructed, so these are explorer-index observations rather than a complete peer-level stale-block census.

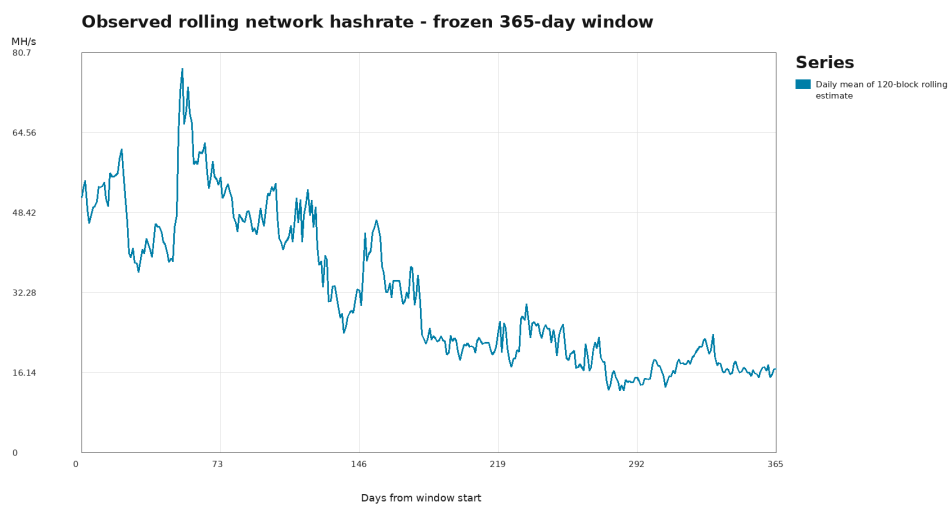


Figure 10: Observed rolling network hashrate over the frozen 365-day window

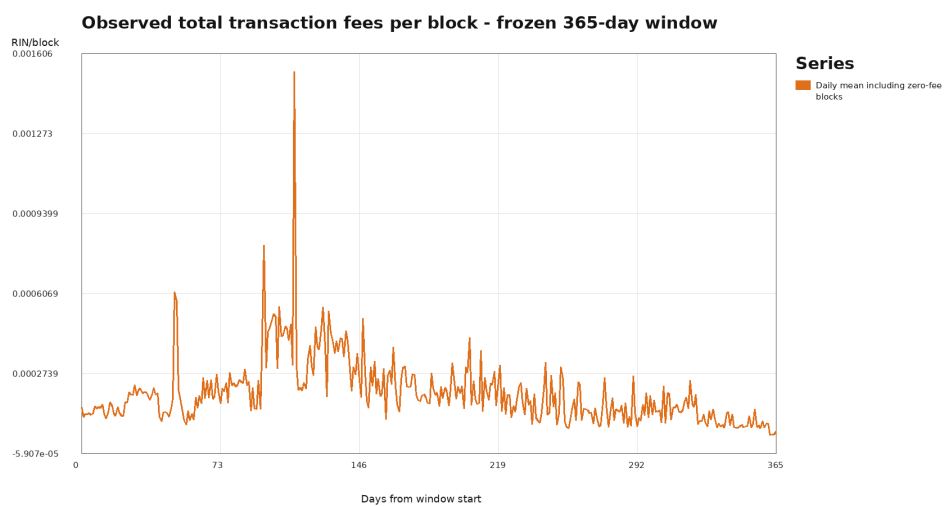


Figure 11: Observed mean total fees per block over the frozen 365-day window

8.4 Historical Price, Volume and Halving Events

CoinPaprika’s free API returned 333 RIN daily historical ticks between 2025-07-28 and 2026-07-27. It rejected the longer request with HTTP 402; that response is frozen with the successful dataset. Missing dates are not interpolated. BTC/USD ticks from the same API are used only to derive RIN/BTC where both daily observations exist ([Rincoin Community Forge 2026b](#)).

Measure	Value	Evidence class
Requested daily period	2025-07-28 to 2026-07-27	frozen source definition
Calendar days	365	derived
Days with RIN price	333	directly observed
Missing RIN-price days	32	directly observed
Low/zero-volume days	326	derived from reported volume

The values are historical daily ticks, not asserted exchange closes. Reported volume is mostly thin, and the historical feed does not include venue-level spread, depth or trade count.

Flag	Explicit criteria
high	Price, executed trades, spread and depth are available from the same venue and period; reported volume exceeds 1,000 USD/day; the last meaningful trade is no more than one day old.
medium	Price and executed volume exceed 100 USD/day, with at least one direct liquidity control (trade count, spread or depth); no material data gap overlaps the window.
low	A price exists, but volume is positive and at most 100 USD/day, historical spread/depth is absent, or staleness cannot be ruled out.
unusable	An anchor price is missing, reported volume is zero with no executed-trade evidence, or a material data gap spans the event.

None of the halving-price windows qualifies as high or medium confidence.

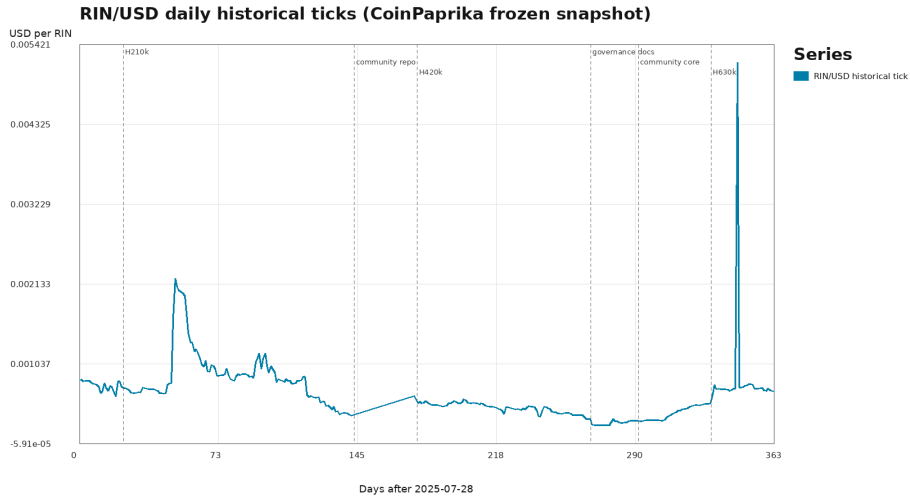


Figure 12: RIN/USD daily historical ticks; vertical markers identify exact or separately sourced project events and do not imply causality

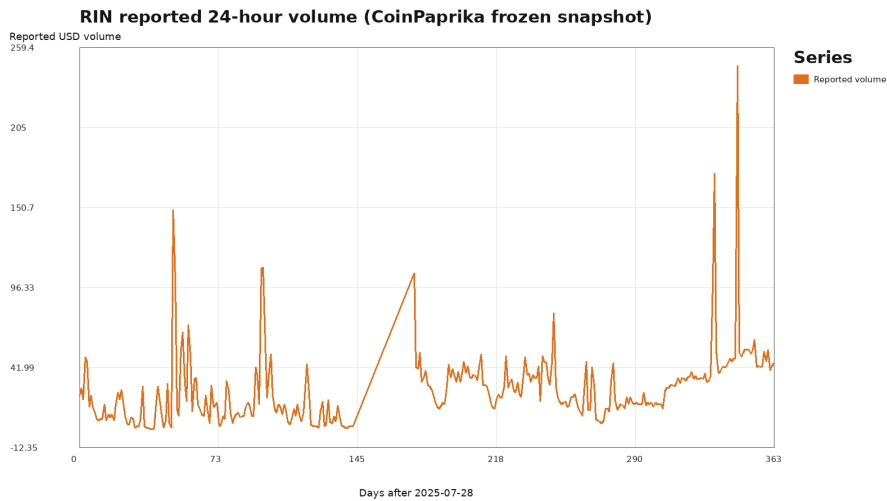


Figure 13: CoinPaprika-reported RIN 24-hour volume; a displayed price can be stale when trading is sparse

Exact main-chain halving times were read from the synchronised node rather than estimated from the 60-second target:

Height	UTC timestamp	Old RIN	New RIN	Required price growth
210,000	2025-08-20T01:52:39Z	50.00	25.00	100%
420,000	2026-01-21T15:54:35Z	25.00	12.50	100%

Height	UTC timestamp	Old RIN	New RIN	Required price growth
630,000	2026-06-24T21:32:43Z	12.50	6.25	100%

A binary halving requires price to double merely to preserve nominal subsidy revenue, with fees and costs held constant. The observed seven-day price windows were:

Height	Pre to event	Event to post	Pre to post	Needed	Confidence
210,000	-4.1096%	-8.7143%	-12.4658%	100%	low
420,000	n/a	-4.6371%	n/a	100%	unusable
630,000	33.5484%	10.6280%	47.7419%	100%	low

At block 210,000, RIN/USD fell over the one- and seven-day symmetric windows; the 30-day pre-event anchor is unavailable. At block 420,000, only the one-day window has complete anchors because CoinPaprika’s RIN series has a material gap. At block 630,000, the one- and seven-day symmetric gains remained below the required 100%. The 30-day symmetric gain exceeded 100%, but most of that movement occurred before the boundary, reported volume was thin, and overlapping market and project events make attribution impossible.

The corresponding 30-day chain windows show:

Height	Hash pre	Hash post	Hash change	Block interval pre/post	Interpretability
210,000	52.64131899	41.89328829	- 20.42%	43.0 / 43.0 s	materially confounded
420,000	35.54309324	21.92447997	- 38.32%	42.0 / 40.0 s	not economically interpretable
630,000	18.23498707	16.45457044	-9.76%	39.0 / 40.0 s	materially confounded

Median rolling hashrate decreased after all three boundaries, by approximately 20.4%, 38.3% and 9.8% respectively. Median observed header-time intervals were broadly stable across the windows, consistent with difficulty response and continued block production. These descriptive changes do not isolate subsidy elasticity. The first event overlaps an operator-reported exchange interruption; the second lacks sufficient price coverage; the third occurs in a thin, changing market. `data/halving_price_hashrate_event_study.csv` therefore leaves precise

RinHash reward-elasticity estimates blank rather than presenting unstable ratios as empirical parameters.

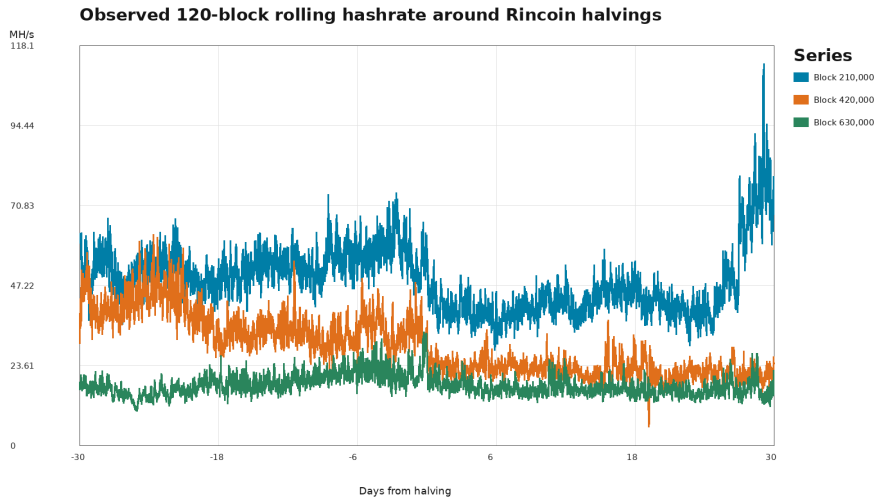


Figure 14: Observed 120-block rolling hashrate in the 30 days around each halving

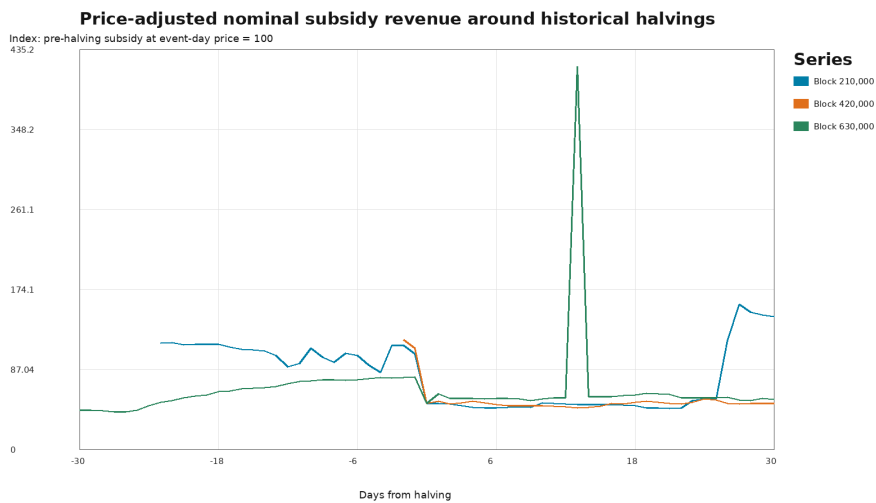


Figure 15: Price-adjusted nominal subsidy revenue around the historical halvings; gaps are retained rather than interpolated

8.5 Market-Event Timeline and Confidence Controls

`data/market_event_timeline.csv` separates exact chain and GitHub dates from operator-supplied venue history. It records the Exbitron listing, reported interruption and eventual RIN-market closure, the shift toward NestEx, the three halvings, governance-document publication, and public Community Core work.

These are neutral event markers. Price movements overlap several events, but low volume, missing venue histories and simultaneous changes prevent causal attribution; price is not treated as a vote by network participants ([Rincoin Community Forge 2026b](#); [Rincoin Community Forge operators 2026](#)).

Date/window	Event	Evidence	Confidence
2025-04-23	Exbitron RIN/USDT listing announcement	directly observed from a public forum source	medium
2025-06-02	CoinPaprika RIN market-data listing announcement	directly observed from a public forum source	medium
summer 2025	Reported Exbitron operational interruption or security incident	operator-supplied evidence	low
2025-08-20T01:52:39Z	First Rincoin binary subsidy halving, block 210,000	directly observed	high
late 2025 to early 2026	Reported transfer of primary RIN trading activity from Exbitron to NestEx	operator-supplied evidence	low
2025-12-19	Public takologi/rincoin repository created	directly observed from frozen GitHub API metadata	high
2026-01-21T15:54:35Z	Second Rincoin binary subsidy halving, block 420,000	directly observed	high
spring 2026	Reported permanent end of RIN trading on Exbitron	operator-supplied evidence	low
2026-04-22	Governance-document repository first published	directly observed from frozen GitHub API metadata	high
2026-05-11	Governance-document set expanded	directly observed from frozen GitHub API metadata	high
2026-05-17	Rincoin Community Core maintenance work publicly committed	directly observed from frozen GitHub API metadata	high
2026-05-21 to 2026-05-22	Governance, fork-status and canonical-source documents expanded	directly observed from frozen GitHub API metadata	high
2026-06-24T21:32:43Z	Third Rincoin binary subsidy halving, block 630,000	directly observed	high

Date/window	Event	Evidence	Confidence
2026-07-27	NestEx is the dominant CoinPaprika-reported RIN venue; Rabid Rabbit and AtomicDEX/Gleec activity is small	directly observed from a frozen public source	medium

CoinPaprika's frozen current market list attributes all reported 24-hour RIN volume to NestEx. Its Gleec Wallet RIN/KMD and RIN/LTC rows reported zero volume, so no AtomicDEX volume is assumed or double counted. Rabid Rabbit remains a separate observation. A public historical order-book series was not available, so old displayed prices cannot be certified as executable.

8.6 RinHash Rental-Market Exposure

Metric	Value	Unit	Evidence status
Visible listed rigs	22	rigs	directly observed
Available and API-online rigs	22	rigs	directly observed status
Advertised available capacity	10.346300	MH/s	quoted capacity; delivery unverified
Observed five-minute delivered rate	0.000000	MH/s	API sample; idle-on-rent rigs may report zero
Lowest ask	0.00001514	BTC/MH/day	quoted ask, not market-clearing price
Capacity / observed honest hashrate	52.50	%	derived from frozen snapshots
Capacity / combined hashrate	34.43	%	derived from frozen snapshots
Recent executed rentals	none reported in 30-day market panel		ask prices are not market-clearing evidence

Snapshot	UTC	Rigs	Available MH/s	Lowest ask	Use
network-aligned base snapshot	2026-07-27T12:30:57Z	22	10.3463	0.00001514	used with the contemporaneous 19.70723840 MH/s node estimate

Snapshot	UTC	Rigs	Available		Use
			MH/s	Lowest ask	
later same-day snapshot	2026-07- 27T16:28:45Z	14	6.6583	0.00001791	reported as a later market observation; not mixed into the aligned baseline

The network-aligned Mining Rig Rentals snapshot contained 22 available, API-online listings advertising 10.3463 MH/s. It was captured with the 19.70723840 MH/s node estimate and is therefore used for the internally aligned rental-to-network ratios. A second snapshot about four hours later contained 14 available listings advertising 6.6583 MH/s. This within-day change demonstrates that visible rental depth is volatile; the aligned value is a dated analytical baseline, not a persistent capacity estimate. Mixing the later rental snapshot with the earlier node estimate would create a false point-in-time comparison.

Both listing sets, their endpoint metadata, and their response hashes are preserved in the frozen evidence snapshots ([Rincoin Community Forge 2026a](#), [2026b](#)).

The listings' five-minute delivered rate was zero at the aligned capture time; several listings state that they start mining when rented. Advertised capacity therefore cannot be treated as proven delivered hashrate.

The market page reported zero rentals and zero unique renters in the preceding 30 days. Quoted asks are consequently not observed market-clearing prices. The capacity-weighted ask was 0.000034110864 BTC/MH/day. Cost calculations use the full visible ask curve rather than applying the cheapest ask to all capacity.

8.6.1 Listing-level rental-cost audit

Rig	Status	Adver- tised MH/s	Ask	Price unit	Mini- mum hours	Minimum BTC
353346	available	0.115000	0.000015140000	BTC/MH/day	24.00	0.000001741100
353213	available	0.129000	0.000017910000	BTC/MH/day	24.00	0.000002310390
353221	available	0.043000	0.000019530000	BTC/MH/day	29.00	0.000001014746
352021	available	0.687500	0.000019770000	BTC/MH/day	3.00	0.000001698984
353312	available	0.073000	0.000021210000	BTC/MH/day	24.00	0.000001548330
353341	available	0.073900	0.000023800000	BTC/MH/day	24.00	0.000001758820
368062	available	0.178000	0.000026110000	BTC/MH/day	6.00	0.000001161895
366932	available	0.046900	0.000033900000	BTC/MH/day	24.00	0.000001589910
351991	available	0.687500	0.000034190000	BTC/MH/day	3.00	0.000002938203
351989	available	0.718750	0.000034960000	BTC/MH/day	3.00	0.000003140938
351994	available	0.687500	0.000035120000	BTC/MH/day	3.00	0.000003018125
351990	available	0.687500	0.000035560000	BTC/MH/day	3.00	0.000003055938
351992	available	0.687500	0.000035880000	BTC/MH/day	3.00	0.000003083438
351995	available	0.718750	0.000035930000	BTC/MH/day	3.00	0.000003228086
357309	available	0.550000	0.000036360000	BTC/MH/day	12.00	0.000009999000
352523	available	0.550000	0.000036360000	BTC/MH/day	12.00	0.000009999000

Rig	Status	Adver- tised MH/s	Ask	Price unit	Mini- mum hours	Minimum BTC
352524	available	0.550000	0.000036360000	BTC/MH/day	12.00	0.000009999000
352520	available	0.550000	0.000036360000	BTC/MH/day	12.00	0.000009999000
352521	available	0.550000	0.000036360000	BTC/MH/day	12.00	0.000009999000
351993	available	0.687500	0.000036930000	BTC/MH/day	3.00	0.000003173672
351984	available	0.687500	0.000037130000	BTC/MH/day	3.00	0.000003190859
351985	available	0.687500	0.000038210000	BTC/MH/day	3.00	0.000003283672

The listing ask is quoted in BTC per MH per day. Each listing's minimum contractual price is therefore:

$$C_{\min,i} = H_i P_i \frac{T_{\min,i}}{24},$$

where H_i is advertised MH/s, P_i is BTC/MH/day, and $T_{\min,i}$ is the minimum rental duration in hours.

Metric	Value	Unit	Calculation
advertised available capacity	10.346300	MH/s	sum of advertised hashrate across available API-online listings
capacity-weighted ask	0.000034110864	BTC/MH/day	sum(listing hashrate * ask) / sum(listing hashrate)
all-capacity weighted 24-hour price	0.000352921232	BTC	sum(listing hashrate * ask * 24/24)
sum of listing minimum contractual prices	0.000090932105	BTC	sum(listing hashrate * ask * listing minimum hours / 24)
minimum-price share of 24-hour price	25.765552	%	sum of listing minimum prices / all-capacity 24-hour price
listings with minimum below / equal to / above 24 hours	16 / 5 / 1	listings	direct classification of listing minimum- duration fields

The sum of listing minimum contract prices differs from the weighted 24-hour price because they answer different questions. The 24-hour value prices every

listed rig for a full day. The contractual minimum prices each rig only for its own minimum duration: 16 of the 22 listings allowed less than 24 hours, five required 24 hours, and one required 29 hours. The reconciled minimum all-capacity ask is therefore 0.000090932105 BTC, while a full 24-hour rental of all capacity is 0.000352921232 BTC. Both are quoted asks; neither establishes delivered hashrate or an executed market price.

Blocks	Hours	Prorated BTC	Minimum- lease BTC	Minimum- lease USD
6	0.1000	0.000001470505	0.000090932105	5.906049
24	0.4000	0.000005882021	0.000090932105	5.906049
144	2.4000	0.000035292123	0.000090932105	5.906049

The theoretical 6-, 24- and 144-block costs prorate asks over 0.1, 0.4 and 2.4 target hours. Actual contracts are constrained by each listing’s minimum, so all three short analytical horizons have the same minimum contractual all-capacity ask of approximately 0.000090932105 BTC, or 5.91 USD at the frozen Kraken XBT/USD price. This remains an ask-based contractual estimate, not total attack cost.

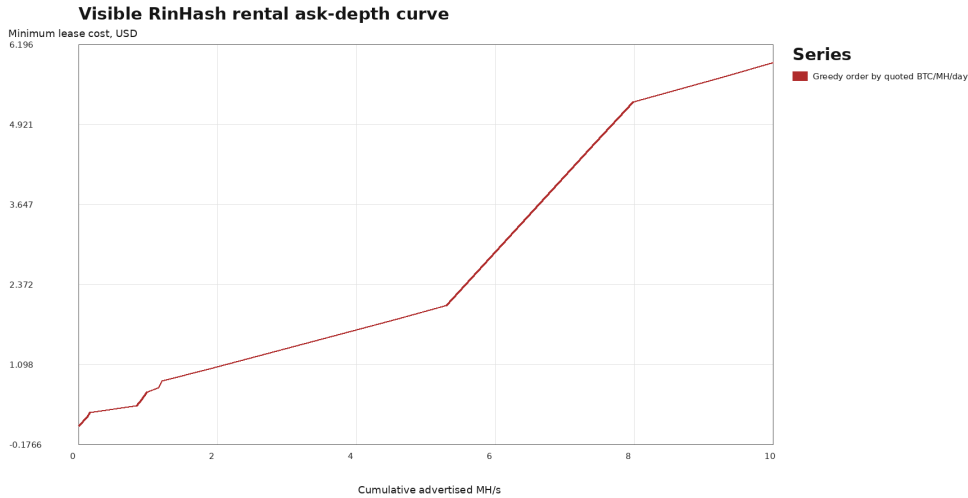


Figure 16: Visible RinHash rental ask-depth curve

A controlled rental validation would be useful but is outside this passive snapshot. It should direct a small legitimate rental to an ordinary mining or test pool, without withholding blocks or attempting a reorganisation, and record setup delay, delivered hashrate, accepted/rejected/stale shares, xnonce behaviour and pool compatibility. Results should be published separately from the quoted market snapshot.

8.7 Miner Hardware Economics

Status	Available evidence	Required next step
Pending	MRR listings expose names and advertised hashrate, but not independently measured wall power or delivered rate.	Execute and publish the controlled benchmark protocol.

No sufficient independently reproducible RinHash dataset was found that combines hardware identity, miner version, measured hashrate and wall power. MRR rig names and advertised hashrates do not supply wall-power measurements and are not independent benchmarks. `data/hardware_benchmark_protocol.csv` therefore specifies a reproducible measurement procedure rather than presenting placeholder CPU, GPU or FPGA classes as empirical evidence.

Until measurements exist, hashprice, electricity break-even, hash per watt and hardware acquisition economics cannot be reported as observed RinHash values. The normalized hardware classes used later are dimensionless sensitivities only.

8.8 Reward-Boundary Shocks and Price Compensation

Scenario	Immediate miner-revenue shock	Price growth needed	Nominal ordinary reduction	Cadence	Terminal
S0	50.00000000%	100.00000000%	50.00%	210,000-block binary integer halvings	zero at 6,930,000
S1	5.00000000%	5.26315789%	5.00%	210,000-block recursive 19/20 reductions	zero at 73,710,000
S2/a	5.00000000%	5.26315789%	5.00%	210,000-block recursive 19/20 reductions	zero at 261,919,612
S2/b	5.00000000%	5.26315789%	5.00%	210,000-block recursive 19/20 reductions	zero at 135,706,704
S3	5.00000000%	5.26315789%	5.00%	210,000-block recursive 19/20 reductions	unbounded tail
S4	0.87719312%	0.88495589%	0.88%	210,000-block 113/114 reductions	zero at 384,300,000
S5/a	50.00000000%	100.00000000%	50.00%	2,100,000-block binary integer halvings	zero at 61,740,000
S5/b	0.00000000%	0.00000000%	50.00%	2,100,000-block binary integer halvings	zero at 63,630,000
S6/a	36.00000000%	56.25000000%	phase-dependent	configured phase boundaries	zero at 280,245,000
S6/b	36.00000000%	56.25000000%	phase-dependent	configured phase boundaries	zero at 234,587,500
S6/c	36.00000000%	56.25000000%	phase-dependent	configured phase boundaries	zero at 233,187,500

Scenario	Immediate miner-revenue shock	Price growth needed	Nominal ordinary reduction	Cadence	Terminal
S7	36.00000000%	56.25000000%	phase-dependent	configured phase boundaries	unbounded tail

Three distinct risks must not be collapsed into one term:

1. **Immediate miner-revenue shock** is the subsidy reduction at height 840,000, holding fees and price constant.
2. **Consensus activation divergence** begins at height 840,000 for every proposed rule because candidate and legacy nodes permit different maximum coinbase values.
3. **First later scheduled reward shock** is the first subsidy reduction after activation.

S0 and S5/a impose a 50% immediate miner-revenue shock and require a 100% price increase for price-only compensation. S1 imposes a 5% immediate shock, creates consensus divergence at height 840,000, and continues with 5% reductions each epoch; price-only compensation for each ideal reduction is approximately 5.263%. S5/b also creates consensus divergence at height 840,000 because legacy software permits only 3.125 RIN while S5/b permits 6.25 RIN. It has **no immediate miner-revenue shock**, but its first later scheduled reward shock is 50% at height 2,730,000. S4 begins with a 0.877% cut and requires approximately 0.885% growth. S6 and S7 begin with a 36% cut from 6.25 to 4 RIN and require 56.25% price growth.

Ordinary reduction rates and terminal integer effects are also reported separately. A percentage calculated from a few remaining base units is not economically comparable to a 50% reduction from several RIN:

Sce-nario	Normal reduction	Largest later reduction, pre-subsidy ≥ 1 RIN	≥ 0.1 RIN	≥ 0.01 RIN	Terminal transition
S0	50.00000000%	50.00000000%	50.00000000%	50.00004096%	0.00000001 RIN to zero at 6,930,000 (100.00000000%; integer-rounding exhaustion)
S1	5.00000000%	5.00000067%	5.00000641%	5.00008281%	0.00000001 RIN to zero at 73,710,000 (100.00000000%; integer-rounding exhaustion)
S2/a	5.00000000%	5.00000067%	5.00000131%	5.00000131%	0.50000000 RIN to zero at 261,919,612 (100.00000000%; predetermined fixed-phase termination)
S2/b	5.00000000%	5.00000067%	5.00000067%	5.00000067%	1.00000000 RIN to zero at 135,706,704 (100.00000000%; predetermined fixed-phase termination)
S3	5.00000000%	5.00000067%	5.00000131%	5.00000131%	n/a
S4	0.87719298%	0.87719392%	0.87720104%	0.87728723%	0.00000001 RIN to zero at 384,300,000 (100.00000000%; integer-rounding exhaustion)

Sce- nario	Normal reduction	Largest later reduction, pre-subsidy ≥ 1 RIN	≥ 0.1 RIN	≥ 0.01 RIN	Terminal transition
S5/a	50.00000000%	50.00000000%	50.00000000%	50.00004096%	0.00000001 RIN to zero at 61,740,000 (100.00000000%; integer-rounding exhaustion)
S5/b	50.00000000%	50.00000000%	50.00000000%	50.00004096%	0.00000001 RIN to zero at 63,630,000 (100.00000000%; integer-rounding exhaustion)
S6/a	phase- dependent	50.00000000%	50.00000000%	50.00000000%	0.50000000 RIN to zero at 280,245,000 (100.00000000%; predetermined fixed-phase termination)
S6/b	phase- dependent	50.00000000%	50.00000000%	50.00000000%	0.60000000 RIN to zero at 234,587,500 (100.00000000%; integer-rounding exhaustion)
S6/c	phase- dependent	50.00000000%	50.00000000%	50.00000000%	0.60000000 RIN to zero at 233,187,500 (100.00000000%; predetermined fixed-phase termination)
S7	phase- dependent	50.00000000%	50.00000000%	50.00000000%	n/a

A later fixed phase or tail emission cannot prevent miner exit caused by an earlier activation shock. It can only affect incentives once that later phase begins. The full deterministic file `data/price_compensation.csv` computes the price-only threshold for every reward boundary; it does not assume that scarcity produces the required price response.

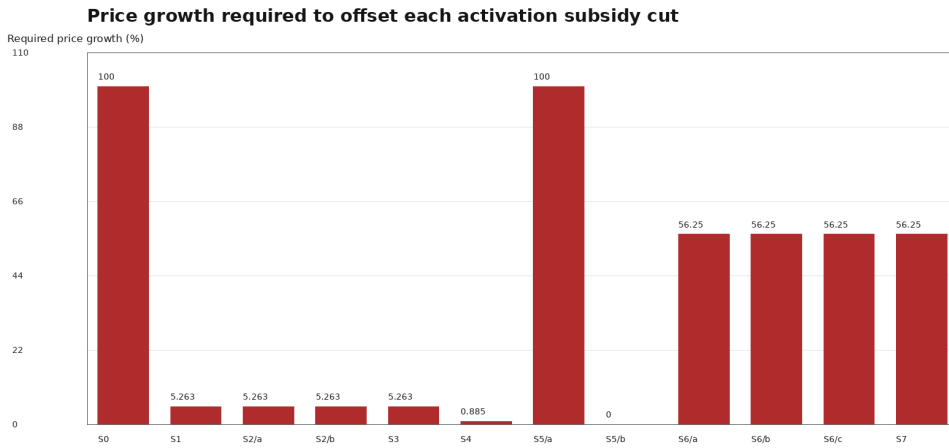


Figure 17: Required price growth to offset each activation subsidy reduction

Dark Gravity Wave v3 adjusts difficulty after block-timing changes. It does not restore revenue lost at a subsidy boundary. Miner exit can therefore produce slow blocks or concentration before difficulty adapts.

8.9 Fee-Market Replacement Requirements

The empirical fee cases use the frozen 90-day distribution. Because 95.14% of blocks had no non-coinbase transaction, the all-block p10, median and p90 fees were all zero. The low case therefore uses the all-block median of zero; the base case uses the all-block mean of 10,002.55 base units, rounded to 10,003; and the high case uses the p90 of 290,698 base units conditional on transaction-bearing blocks. That conditional high case must not be read as an all-block p90.

Scenario	Before	After	Additional fee required	Multiple of 90 d mean	Transac- tions at median fee
S0	6.25000000	3.12500000	3.12500000	31242.0388	3126
S1	6.25000000	5.93750000	0.31250000	3124.2039	313
S2/a	6.25000000	5.93750000	0.31250000	3124.2039	313
S2/b	6.25000000	5.93750000	0.31250000	3124.2039	313
S3	6.25000000	5.93750000	0.31250000	3124.2039	313
S4	6.25000000	6.19517543	0.05482457	548.1060	55
S5/a	6.25000000	3.12500000	3.12500000	31242.0388	3126
S5/b	6.25000000	6.25000000	0.00000000	0.0000	0
S6/a	6.25000000	4.00000000	2.25000000	22494.2679	2251
S6/b	6.25000000	4.00000000	2.25000000	22494.2679	2251
S6/c	6.25000000	4.00000000	2.25000000	22494.2679	2251
S7	6.25000000	4.00000000	2.25000000	22494.2679	2251

At activation, replacing a 5% S1/S2/S3 boundary would require an additional 0.3125 RIN per block, about 3,124 times the observed 90-day all-block mean fee and about 313 transactions at the observed median fee per transaction. Replacing the S0/S5/a 50% boundary would require 3.125 RIN per block, about 31,242 times the mean and 3,126 median-fee transactions per block. S5/b requires no fee replacement at activation, but its first halving at height 2,730,000 creates the same 3.125 RIN-per-block gap if fees and subsidy remain otherwise unchanged. Even S4's smaller first reduction would require roughly 548 times the observed mean fee.

These ratios show that fee replacement for every non-zero modelled immediate subsidy shock is not remotely supported by current usage. S5/b requires no fee replacement at activation, but would face a later 3.125 RIN-per-block subsidy gap at height 2,730,000 under unchanged fees and price. These results do not predict future demand, fee policy or price. The full file includes fee share before and after every boundary, both with unchanged observed fees and with complete hypothetical fee replacement.

8.9.1 Transaction value is not transaction byte size

The frozen dataset contains serialization and input/output counts for the same 68,946 transactions used for exact fee reconstruction. The earlier scatter plot has been removed: it connected a one-in-twenty sample as a line, mixed different transaction structures, compressed dense observations and extreme outliers, and did not communicate a defensible relationship.

Specifically, the removed graphic placed $\log_{10}(\text{aggregate output RIN})$ on a linear x-axis and serialized bytes on a linear y-axis. Aggregate outputs included possible change. A dense 222-byte cluster, 1,392 transactions over 10,000 bytes, 19 over 100,000 bytes, and a 217,303-byte maximum made the sampled connected trace visually uninformative. Its caption did not explain the one-in-twenty sampling or structural mixture. The intended question is instead whether transferred economic value determines serialized size; the available data support a structural comparison, not that causal claim.

The most common exact input/output structures show why structure must be considered explicitly:

In-puts	Out-puts	Transac-tions	Size p10	Median size	Size p90	Median aggregate output
1	2	39366	222 B	222 B	222 B	4026.24207808 RIN
2	2	1982	366 B	369 B	370 B	24.99970600 RIN
1	3	1652	253 B	253 B	256 B	12.49998250 RIN
2	3	1208	397 B	400 B	404 B	25.00040998 RIN
3	2	698	515 B	516 B	518 B	37.49930796 RIN
1	1	685	191 B	191 B	191 B	67.39491290 RIN
1	4	671	284 B	284 B	287 B	12.49959241 RIN
2	4	595	428 B	432 B	435 B	24.99945694 RIN
2	5	579	463 B	463 B	463 B	24.99939438 RIN
3	3	567	544 B	547 B	552 B	37.49938220 RIN

For example, the 39,366 observed one-input/two-output transactions have a median serialized size of 222 bytes and a p10-p90 range of 222-222 bytes even though their aggregate output values vary. Across the complete dataset, the descriptive associations are:

Variables	Measure	Value	Limit
$\log_{10}$ aggregate output RIN vs $\log_{10}$ serialized bytes	Pearson correlation	-0.16140191	descriptive association only; mixed structures and known test activity

Variables	Measure	Value	Limit
aggregate output RIN rank vs serialized-byte rank	Spearman rank correlation	-0.27891827	descriptive association only; aggregate outputs include possible change
input-plus-output count vs serialized bytes	Pearson correlation	0.99833884	structure proxy; script and witness composition also affect size
input-plus-output-count rank vs serialized-byte rank	Spearman rank correlation	0.96322307	structure proxy; no causal estimate is claimed

The strong association between input-plus-output count and serialized size is a structural diagnostic, not a causal estimate; script types, witness data and meta-data also affect serialization. Conversely, the weak negative association between aggregate output value and size does not show that higher-value transactions are smaller. It reflects a mixed, non-random set of transaction structures and activity.

Aggregate output value includes possible change and is not a payment-value estimate. The observation interval is known to contain maintainer test transactions. The frozen evidence does not provide approved address or time labels with which to remove them, so they remain in the calculations and confound economic-value interpretation. They do not alter the mechanical fact that serialized size is determined by transaction structure and encoding, not by the number of RIN transferred. The binned results are preserved in `data/transaction_size_by_value_range.csv`; no causality is inferred from any correlation.

8.10 Price and Liquidity Sensitivity

Venue	Best bid	Best ask	Spread, %	Bid depth within 5%, RIN	Reported 24 h volume, RIN	9,000 RIN sale-walk average
NestEx	0.00064861	0.00066172	2.00	18440.91901400	69867.09449702	0.00064428 (complete)
Rabid Rabbit	0.00006400	0.00099980	175.94	23689.51612903	unavailable	0.00006400 (complete)

NestEx and Rabid Rabbit are retained as separate observations because their prices, spreads and available metadata differ materially. At NestEx, visible bids within 5% totalled about 18,440.9 RIN. Walking the scheduled 9,000 RIN daily subsidy through the frozen bids completed at an average 0.000644281769

USDT/RIN, 0.667% below the best bid. Scheduled daily issuance was 12.88% of NestEx’s reported 24-hour RIN volume. This is one static order-book walk, not an estimate of recurring daily impact.

Rabid Rabbit showed a much wider spread and did not expose compatible 24-hour volume through the frozen endpoint. Its values are therefore not merged with NestEx. Exchange depth can change faster than block-level monetary data, and displayed orders may be cancelled or may not represent executable capacity.

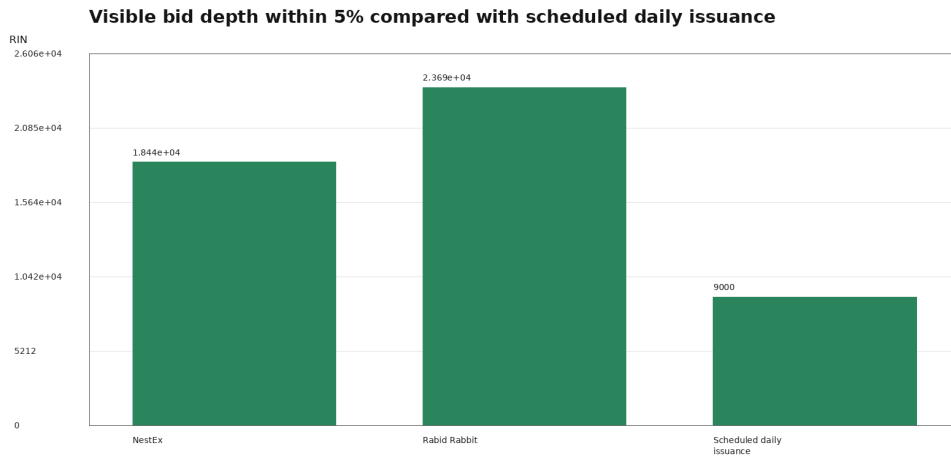


Figure 18: Visible bid depth compared with scheduled daily issuance

8.10.1 NestEx liquidity pool

Snapshot	Base label/value	USDT	Total USD	Largest LP	Evidence
operator-supplied screenshot	Pooled NEX: 450742.4413250	296.209619	592.410	84.51%	operator-supplied evidence
public RIN liquidity endpoint	Pooled RIN: 450546.550872	296.340079	592.68	84.52%	evidence directly observed from a frozen public source

The operator-supplied screenshot labels its base quantity **Pooled NEX** but there should be **RIN**. This was a flaw which was reported and fixed by the NestEx staff immediately. A separately frozen public RIN endpoint collected the same day explicitly labels a near-identical quantity **Pooled RIN**. This supports an interpretation that the earlier interface label was inconsistent, but the source record is not silently rewritten.

NestEx describes the facility as an exchange-managed automated liquidity reserve integrated with its centralised order book. The analysis therefore measures executable depth from the order book and does not apply a generic constant-product slippage formula. Total displayed LP capital is not equivalent to executable bid depth within 5%.

The public endpoint reported three providers. The largest position was 84.52%, HHI was approximately 0.7300, and the effective provider count was about 1.37. The identity of the largest provider as Rincoin Community is operator-supplied, not independently verified.

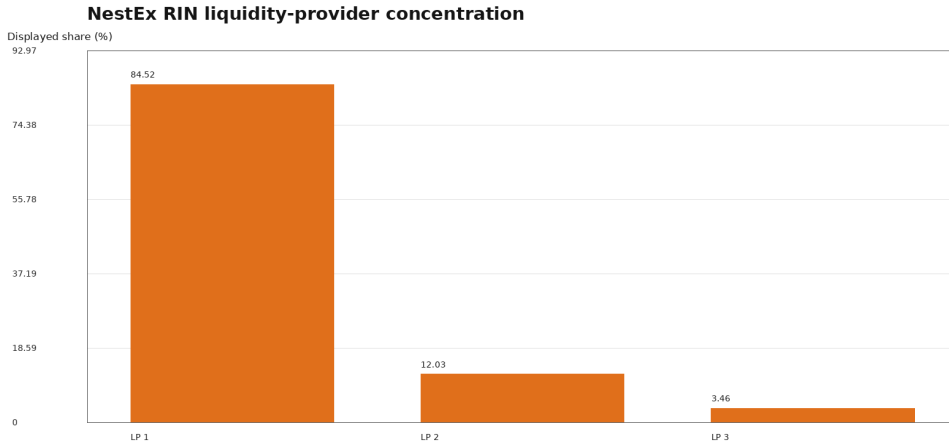


Figure 19: Displayed NestEx RIN liquidity-provider concentration

Sixty rate-limited public tradebook pages provide a frozen executed-trade history in `data/nestex_trade_history.csv`. The repeated order-book/LP collector begins with one dated observation; it does not yet constitute a 14- or 30-day depth series.

8.11 Sticky and Price-Elastic Hashrate

The homogeneous exit model is replaced by:

$$H_{\text{honest}} = H_{\text{committed}} + H_{\text{elastic}}(1 - e),$$

where (e) is applied only to elastic hash. The frozen network estimate is 19.70723840 MH/s. At least 4 MH/s, or approximately 20.3%, is reported by the operator as Rincoin Community infrastructure and is not listed on Mining Rig Rentals. It is described as **operator-committed under current policy**,

not permanently price-independent. This 4 MH/s value is operator-supplied rather than independently verified ([Rincoin Community Forge operators 2026](#)).

Elastic exit	Honest MH/s	Rental / honest	Rental / combined	Rental exceeds honest
0.00%	19.70723840	52.49999919%	34.42622916%	false
10.00%	18.13651456	57.04679345%	36.32471074%	false
25.00%	15.78042880	65.56412459%	39.60044167%	false
50.00%	11.85361920	87.28389048%	46.60512458%	false
75.00%	7.92680960	130.52287770%	56.62035760%	true

With 4 MH/s committed and no attacker-owned hash, visible rental capacity does not exceed remaining honest hash through the 50% elastic-exit case. It exceeds remaining honest hash in the 75% case. Without committed hash, visible rental capacity exceeds the modelled honest network after a 50% elastic exit. `data/sticky_elastic_hashrate_scenarios.csv` includes committed-hash cases of 0, 2 and 4 MH/s, elastic exits of 0%, 10%, 25%, 50% and 75%, and configurable attacker-owned hash of 0, 2, 5 and 10 MH/s.

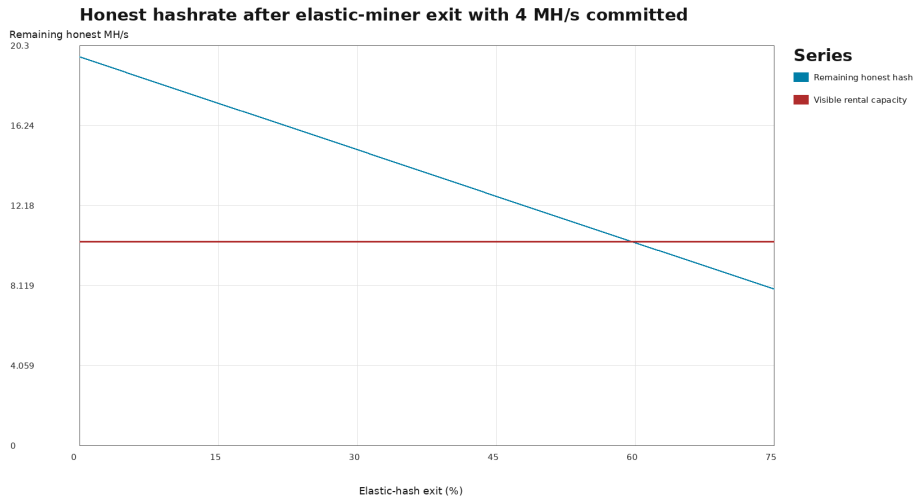


Figure 20: Remaining honest hashrate under the 4 MH/s operator-committed case

Committed infrastructure improves downside resilience, but it also creates dependency. The same operator is reported to supply approximately 4 MH/s, about 85% of displayed NestEx LP value, Fulcrum and other services. Resources controlled by one operator are not counted as independent redundancy. `data/cross_layer_operator_concentration.csv` separates the resilience contribution from this concentration risk ([Rincoin Community Forge operators 2026](#)).

8.12 Attack-Capacity and Reorganisation-Risk Proxies

Honest-hash reduction	Honest MH/s	Rental / honest	Rental / combined	Rental alone exceeds honest?	Min 24 h ask cost if sufficient
0%	19.707238	52.5000%	34.4262%	false	insufficient visible depth
10%	17.736515	58.3333%	36.8421%	false	insufficient visible depth
25%	14.780429	70.0000%	41.1765%	false	insufficient visible depth
50%	9.853619	105.0000%	51.2195%	true	\$5.906049

The additive zero-overlap case assumes none of the visible MRR capacity is already contributing to the observed network. Under that assumption, visible rental capacity equalled 52.5% of pre-existing honest hashrate but 34.43% of combined honest-plus-rental hashrate, and did not exceed the observed honest network. That conclusion must not be generalized to unknown overlap.

Let (α) be the fraction of visible rental capacity already contributing to observed network hashrate. Redirecting rented capacity then removes (H_r) from the honest chain while adding the full (H_r) to the attacker chain. The operator-supplied 4 MH/s Community Forge component is not listed on MRR and remains outside this overlap.

Alpha	Removed MH/s	Remaining honest	Attacker MH/s	Attacker / combined	Attacker exceeds honest
0.00%	0.00000000	19.70723840	10.34630000	34.42622916%	false
25.00%	2.58657500	17.12066340	10.34630000	37.66816102%	false
50.00%	5.17315000	14.53408840	10.34630000	41.58415791%	false
75.00%	7.75972500	11.94751340	10.34630000	46.40883914%	false
100.00%	10.34630000	9.36093840	10.34630000	52.49999919%	true

At 100% overlap, the visible rental capacity exceeds the remaining honest hashrate in this static model. The overlap is not observed, so the table is a speculative stress range rather than an estimate of actual attacker control. It corrects the stronger additive-only statement: visible rental capacity is insufficient only in the zero-overlap case and the lower modelled overlap cases.

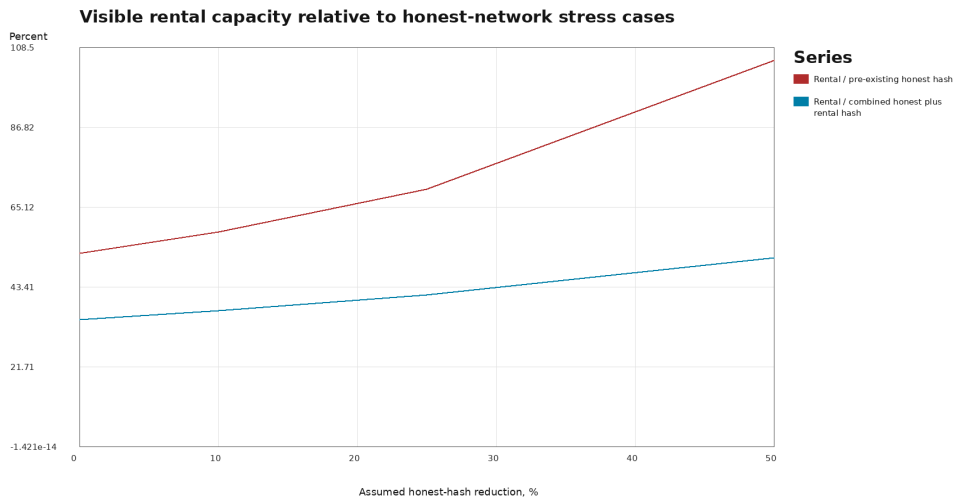


Figure 21: Visible rental capacity relative to honest-network stress cases

`data/rental_attack_capacity.csv` walks the quoted ask curve and respects each rig’s minimum rental duration. The sticky/elastic file combines the same visible rental capacity with configurable owned hash. Both report whether visible depth is sufficient rather than extrapolating beyond listed rigs.

Rental expense is not total attack cost. The analysis does not observe private hashrate, delivered rental reliability, setup delay, pool rejection, detection and countermeasures, exchange confirmation changes, reorganisation success probability, double-spend liquidity or price collapse. The observed explorer orphans were depth 1; this does not establish a safe confirmation count or bound deeper adversarial reorganisations.

8.12.1 Confirmation policies and mathematical catch-up proxy

The following service settings are operator-supplied evidence. They describe different contexts and are not interchangeable security standards ([Rincoin Community Forge operators 2026](#)).

Service	Confir- ma- tions	Policy context	Evidence
Rincoin Core convention	6	wallet display convention	operator-supplied evidence
Electrin	6	wallet display convention	operator-supplied evidence
NestEx deposits	20	custodial exchange deposit policy	operator-supplied evidence

Service	Confir- ma- tions	Policy context	Evidence
AtomicDEX setting	2	atomic-swap protocol setting	operator-supplied evidence

AtomicDEX’s two-confirmation value is an atomic-swap protocol setting, not a custodial exchange deposit policy. Rincoin Core and Electrin values describe wallet-display conventions; NestEx’s value is a custodial deposit policy.

For comparison only, the next table applies the elementary biased-random-walk eventual catch-up expression (1) for attacker fraction (q 0.5), and $((q/(1-q))^z)$ otherwise, at confirmation depth (z). It uses each overlap case’s static attacker share from the preceding table.

Alpha	2 conf	6 conf	12 conf	20 conf	24 conf
0.00%	27.562499%	2.093899%	0.043844%	0.000253%	0.000019%
25.00%	36.519848%	4.870649%	0.237232%	0.004220%	0.000563%
50.00%	50.675091%	13.013185%	1.693430%	0.111672%	0.028677%
75.00%	74.992025%	42.174044%	17.786500%	5.625366%	3.163596%
100.00%	100.000000%	100.000000%	100.000000%	100.000000%	100.000000%

These are mathematical proxies, not empirical reorganisation probabilities. They omit finite attack duration, difficulty changes, block-time variance, network propagation, detection, service response, rental delivery, and attacker economics. Their purpose is to show why confirmation depth and rental overlap must be analysed together, not to certify any confirmation policy as safe.

8.13 Scenario Security Comparison

Scenario	Activation subsidy	Block revenue at observed fee	Marginal USDT value	Boundary shock, %	Normal- ized 10 y index
S0	3.12500000	3.12510003	0.00205269	- 50.00000000	47.36076480
S1	5.93750000	5.93760003	0.00390005	- 5.00000000	779144.45363280
S2/a	5.93750000	5.93760003	0.00390005	- 5.00000000	779144.45363280
S2/b	5.93750000	5.93760003	0.00390005	- 5.00000000	779144.45363280

Scenario	Activation subsidy	Block revenue at observed fee	Marginal USDT value	Boundary shock, %	Normal- ized 10 y index
S3	5.93750000	5.93760003	0.00390005	- 5.00000000	779144.45363280
S4	6.19517543	6.19527546	0.00406930	- 0.87719312	2351256.50739600
S5/a	3.12500000	3.12510003	0.00205269	- 50.00000000	369609.81819120
S5/b	6.25000000	6.25010003	0.00410532	0.00000000	739172.31819120
S6/a	4.00000000	4.00010003	0.00262743	- 36.00000000	473087.31819120
S6/b	4.00000000	4.00010003	0.00262743	- 36.00000000	473087.31819120
S6/c	4.00000000	4.00010003	0.00262743	- 36.00000000	473087.31819120
S7	4.00000000	4.00010003	0.00262743	- 36.00000000	473087.31819120

This table combines deterministic activation subsidies with the frozen 90-day mean fee and NestEx marginal price. Price and fees are held constant solely to expose the boundary arithmetic; the resulting USDT values are not forecasts for block height 840,000. The final column remains a normalized sensitivity under a different common-assumption model.

S0 and S5/a have the largest immediate nominal revenue reduction. S6 and S7 are intermediate, S1/S2/S3 are smaller, S4 is smallest, and S5/b has no subsidy change at the activation boundary. S5/b consequently preserves the highest activation subsidy among the bounded corrective candidates, but its first delayed boundary is still a 50% cut. At later horizons, fixed or perpetual phases maintain more nominal subsidy than finite declining schedules, but their issuance trade-offs remain those reported in the deterministic analysis. A larger subsidy supports a larger gross revenue figure under equal price and fees; it does not establish realised security without hardware, liquidity and miner-response evidence.

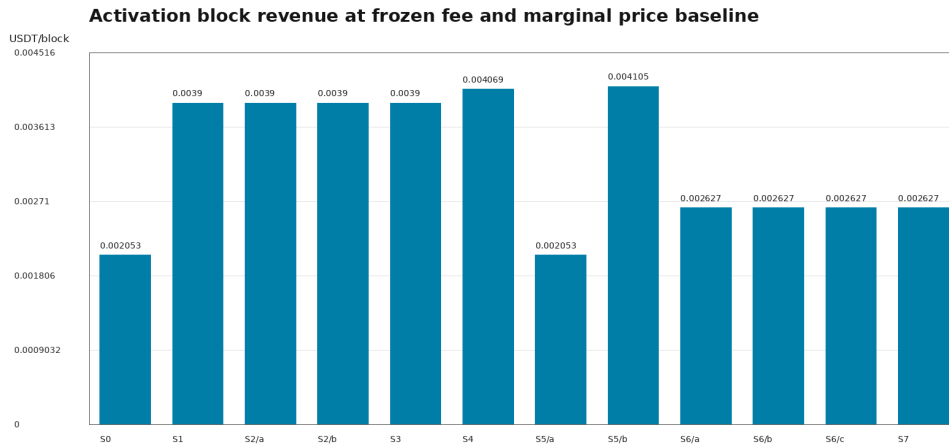


Figure 22: Scenario activation revenue at the frozen fee and marginal-price baseline

The empirical overlay does not change the monetary score and does not produce a synthetic total. It distinguishes immediate miner-revenue shock, consensus activation divergence, first later scheduled reward shock, subsidy runway, and eventual fee dependence. A fixed phase can improve later nominal revenue but does not repair a severe earlier miner-revenue shock. No scenario has evidence that current fees could replace its immediate subsidy loss.

8.14 External Research and Transfer Limits

Topic	Primary source	Rincoin transfer limit
reward and hash-supply elasticity	Kawaguchi, Komiyama and Noda (2026), Miners' Reward Elasticity and Stability of Competing Proof-of-Work Cryptocurrencies (Kawaguchi et al. 2026)	BTC/BCH/BSV estimates cannot be transplanted to RinHash.
price-driven multi-chain hash allocation	Bissias, Levine and Thibodeau (2018), Using Economic Risk to Model Miner Hash Rate Allocation in Cryptocurrencies (Bissias et al. 2018)	RinHash market depth and miner choices differ from SHA-256 markets.
difficulty-adjustment instability	Ilie, Werner, Stewart and Knottenbelt (2020), Unstable Throughput: When the Difficulty Algorithm Breaks (Ilie et al. 2020)	The studied BCH DAA is not Rincoin's DGWv3.
fee-only mining security	Carlsten, Kalodner, Weinberg and Narayanan (2016), On the Instability of Bitcoin Without the Block Reward (Carlsten et al. 2016)	Bitcoin fee demand and block timing are not Rincoin observations.

Topic	Primary source	Rincoin transfer limit
fee-market economics	Easley, O'Hara and Basu (2019), From Mining to Markets: The Evolution of Bitcoin Transaction Fees (Easley et al. 2019)	Bitcoin usage and liquidity are materially different.
majority-attack economics	Budish (2018), The Economic Limits of Bitcoin and the Blockchain (Budish 2018)	A theoretical constraint, not a Rincoin attack-cost estimate.
PoW security and performance	Gervais et al. (2016), On the Security and Performance of Proof of Work Blockchains (Gervais et al. 2016)	Parameters require Rincoin-specific calibration.

Primary research supports analysing reward elasticity, cross-chain allocation, DAA feedback, fee-only incentives and attack economics (Kawaguchi et al. 2026; Bissias et al. 2018; Ilie et al. 2020; Carlsten et al. 2016; Easley et al. 2019; Budish 2018; Gervais et al. 2016). It does not supply RinHash calibration. Published BTC, BCH or BSV elasticities are not transplanted to Rincoin because hardware interchangeability, market depth, DAA rules and miner populations differ.

8.15 Normalised Long-Run Sensitivities

The original normalised model is retained as a secondary scenario-analysis layer. It applies common stress, flat-price and moderate-growth price indices; empirically anchored fee cases; placeholder hardware-cost classes; and explicit liquidity-realisation factors. These are sensitivity controls, not observations or forecasts.

Scenario	Years	Subsidy	Fee share, %	Revenue index	Sustainable-hashrate index
S0	1	0.78125000	0.01280220	369609.81819120	522733.88572755
S0	10	0.00000009	99.91010787	47.36076480	66.98165307
S0	25	0.00000000	100.00000000	47.31819120	66.92144184
S0	50	0.00000000	100.00000000	47.31819120	66.92144184
S1	1	5.35859375	0.00186669	2534876.50569120	3585039.62947755
S1	10	1.64700054	0.00607310	779144.45363280	1101932.87013782
S1	25	0.24686925	0.04050301	116826.34821120	165225.83532727
S1	50	0.00975101	1.01542578	4659.93596160	6590.48085998
S2/a	1	5.35859375	0.00186669	2534876.50569120	3585039.62947755
S2/a	10	1.64700054	0.00607310	779144.45363280	1101932.87013782
S2/a	25	0.50000000	0.02000200	236567.31819120	334573.77858470
S2/a	50	0.50000000	0.02000200	236567.31819120	334573.77858470
S2/b	1	5.35859375	0.00186669	2534876.50569120	3585039.62947755
S2/b	10	1.64700054	0.00607310	779144.45363280	1101932.87013782
S2/b	25	1.00000000	0.01000200	473087.31819120	669080.63572755
S2/b	50	1.00000000	0.01000200	473087.31819120	669080.63572755
S3	1	5.35859375	0.00186669	2534876.50569120	3585039.62947755
S3	10	1.64700054	0.00607310	779144.45363280	1101932.87013782
S3	25	0.50000000	0.02000200	236567.31819120	334573.77858470
S3	50	0.50000000	0.02000200	236567.31819120	334573.77858470
S4	1	6.08696483	0.00164332	2879425.16137440	4072329.87108665
S4	10	4.97042362	0.00201246	2351256.50739600	3325348.48903149
S4	25	3.58770144	0.00278806	1697173.60736880	2400288.38756445
S4	50	2.05945615	0.00485687	974252.45538720	1377871.32976190
S5/a	1	3.12500000	0.00320086	1478297.31819120	2090734.77858470
S5/a	10	0.78125000	0.01280220	369609.81819120	522733.88572755

Scenario	Years	Subsidy	Fee share, %	Revenue index	Sustainable-hashrate index
S5/a	25	0.04882812	0.20444264	23144.97207600	32733.60336463
S5/a	50	0.00076293	11.59149903	408.21459840	577.33207488
S5/b	1	6.25000000	0.00160045	2956547.31819120	4181402.63572755
S5/b	10	1.56250000	0.00640151	739172.31819120	1045400.85001327
S5/b	25	0.09765625	0.10232591	46242.63069120	65400.29197755
S5/b	50	0.00152587	6.15228489	769.11573600	1087.74939806
S6/a	1	4.00000000	0.00250069	1892207.31819120	2676121.77858470
S6/a	10	1.00000000	0.01000200	473087.31819120	669080.63572755
S6/a	25	0.50000000	0.02000200	236567.31819120	334573.77858470
S6/a	50	0.50000000	0.02000200	236567.31819120	334573.77858470
S6/b	1	4.00000000	0.00250069	1892207.31819120	2676121.77858470
S6/b	10	1.00000000	0.01000200	473087.31819120	669080.63572755
S6/b	25	0.60000000	0.01666889	283871.31819120	401475.15001327
S6/b	50	0.60000000	0.01666889	283871.31819120	401475.15001327
S6/c	1	4.00000000	0.00250069	1892207.31819120	2676121.77858470
S6/c	10	1.00000000	0.01000200	473087.31819120	669080.63572755
S6/c	25	0.60000000	0.01666889	283871.31819120	401475.15001327
S6/c	50	0.60000000	0.01666889	283871.31819120	401475.15001327
S7	1	4.00000000	0.00250069	1892207.31819120	2676121.77858470
S7	10	1.00000000	0.01000200	473087.31819120	669080.63572755
S7	25	0.60000000	0.01666889	283871.31819120	401475.15001327
S7	50	0.60000000	0.01666889	283871.31819120	401475.15001327

The table uses the flat-price, observed-base-fee and normalised reference-hardware case. The complete Cartesian output is `data/miner_revenue_break_even.csv`. Its sustainable-hashrate index means revenue-supported normalised cost units; it is not hashes per second.

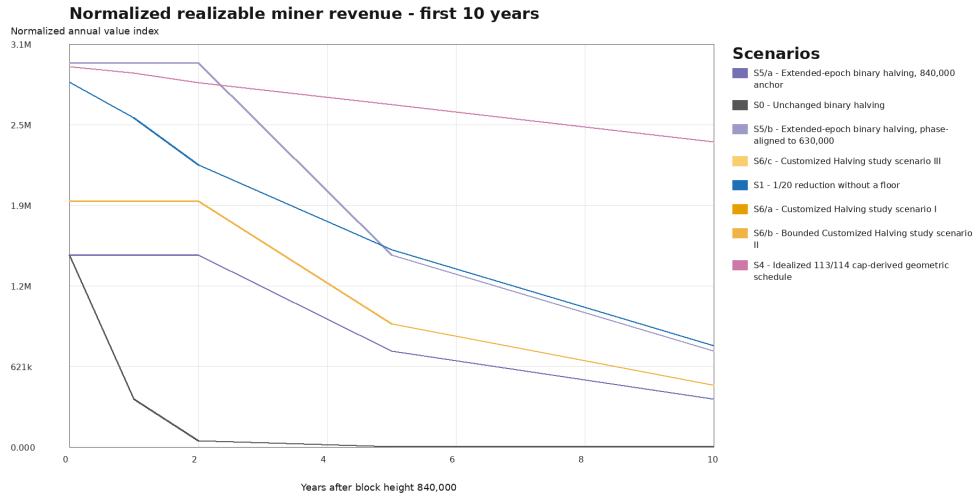


Figure 23: Normalised realisable miner revenue under common assumptions, first 10 years

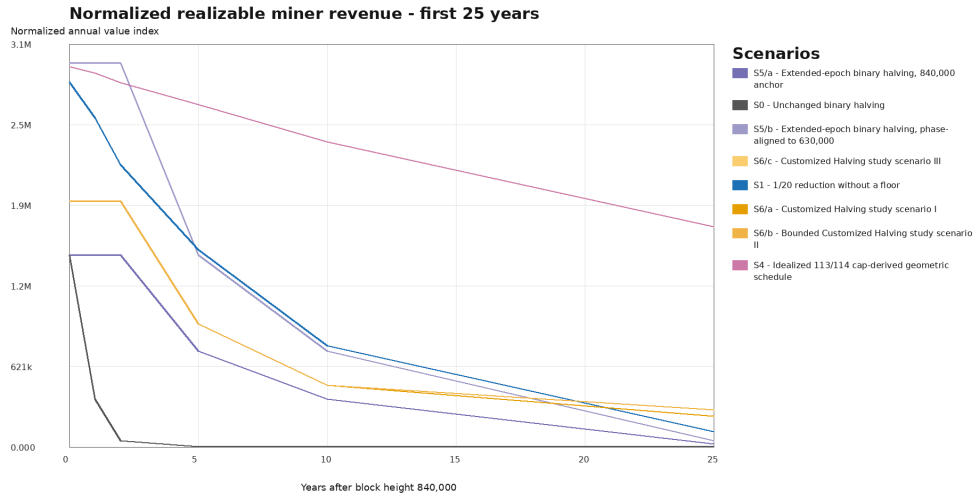


Figure 24: Normalised realisable miner revenue under common assumptions, first 25 years

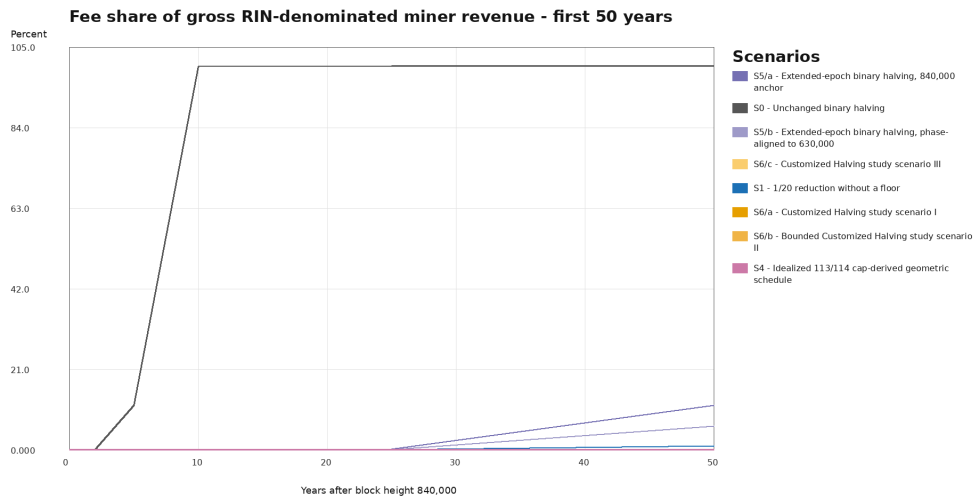


Figure 25: Fee share under common normalised assumptions, first 50 years

Long-run normalised price paths are not empirical forecasts and do not select between the focused candidates. Multi-century price compounding is omitted from the main interpretation because it creates false precision. The assumption matrix and generic attack proxies are documented in the appendix.

8.16 Limitations

The snapshot is dated. Point-in-time market depth can be withdrawn, quoted rental capacity has not been validated by an executed rental, and the repeated

NestEx order-book/LP series has only one observation. CoinPaprika history starts in July 2025 and contains a material gap. Exbitron event dates remain partly operator-supplied. No measured RinHash power-efficiency dataset is available. Pool shares, private hashrate and hardware opportunity cost remain unobserved.

Observed price, fees, volume and hashrate do not define a stable equilibrium. Stress cases do not estimate probability. Rental asks do not establish delivered hashrate or market-clearing cost, and rental cost is not reorganisation cost. Exchange depth does not establish realisable double-spend value. DGWv3 behaviour around a future reward boundary still requires a chain-level transition simulation.

The robust findings are narrower:

1. no historical halving provides a high-confidence case that price promptly and durably doubled to offset the reward cut;
2. observed median hashrate fell in each 30-day post-halving window, but event confounding prevents a causal elasticity estimate;
3. current fee revenue is negligible relative to every non-zero modelled immediate subsidy shock; S5/b has no such shock at activation but has a later 3.125 RIN-per-block subsidy gap at height 2,730,000 under unchanged fees and price;
4. operator-committed hash materially improves the downside stress case while introducing single-operator dependency;
5. visible rental asks are material relative to current hashrate; they do not exceed the observed honest network in the zero-overlap additive case, but exceed remaining honest hash in the 100%-overlap static case and in severe elastic-miner-exit cases;
6. a later fixed phase does not repair an earlier miner-revenue shock; and
7. current liquidity is thin and concentrated, so displayed price and volume cannot be treated as a stable security budget.

9 Conclusions and Public Review

9.1 Preliminary Conclusion

The principal public-review set is **S1, S5/b and S6/b**. S1 and S5/b form a controlled comparison pair: their ideal post-840,000 issuance is exactly equal. Integer consensus arithmetic produces maximum scheduled issuance of 44,624,993.01750000 RIN for S1 and 44,624,999.76900000 RIN for S5/b, a difference of only 6.75150000 RIN. The unresolved choice is therefore about the timing and shape of issuance, not its final amount within that pair.

S1 is the stronger candidate if the principal objective is to eliminate large scheduled reward cliffs and allow miner economics to adjust gradually. It imposes a 5% immediate miner-revenue shock at height 840,000 and repeats 5% reductions every 210,000 blocks. This provides continuous predictability and limits each scheduled adjustment, but creates frequent monetary changes and no rare consensus-level halving milestone.

S5/b is the stronger candidate if the principal objective is to avoid an immediate subsidy reduction, restore a familiar four-year monetary rhythm, and preserve major public halving milestones. For heights at and above activation, its schedule is phase-aligned to height 630,000 without changing historical validation. It keeps 6.25 RIN through height 2,729,999 and imposes its first later 50% reduction at height 2,730,000. The approximately 3.60-year interval offers preparation time, but current evidence does not show that fees or price will absorb that future shock.

Both candidates create validation rules divergence at height 840,000. S1 permits 5.9375 RIN where legacy software permits 3.125 RIN; S5/b permits 6.25 RIN. S5/b therefore has no immediate miner-revenue shock, not an absence of consensus activation risk.

S6/b is the bounded representative of Customized Halving Scenario II. It uses 4 RIN from activation, 2 RIN from height 2,100,000, 1 RIN from height 4,200,000, and 0.6 RIN from height 6,300,000 through 234,587,499. Its precomputed zero-subsidy height is 234,587,500 and its maximum scheduled issuance is exactly 168,000,000 RIN. It prioritizes continuity with that published direction and a centuries-long nominal subsidy. Its trade-offs are a 36% activation reduction, later 50%, 50% and 40% reductions, a roughly 434-year final phase, missing source derivations, and a broader governance contingency excluded from the fixed rule. The inspected S7 implementation is unbounded and is not evidence that this cutoff is implemented.

The current numerical score does not resolve the choice. S1's higher raw score depends partly on a ten-year scheduled-supply threshold that places S5/b just over the 100% boundary. The score also excludes the possible communication and coordination value, and possible volatility cost, of major halving milestones.

These omissions are deliberate because those effects are qualitative and not established by the deterministic model.

S5/a remains a lower-issuance comparison but imposes a 50% immediate miner-revenue shock. S0 remains the no-hard-fork baseline. S2 and S6 illustrate the multi-century consequences of bounded constant phases, S4 is the cap-derived geometric comparator, and S3 and S7 remain unbounded security-budget comparators.

The unresolved policy question is:

Should Rincoin distribute the remaining issuance through smooth recurring reductions, through a familiar sequence of infrequent but highly visible halvings, or through the bounded $4 \rightarrow 2 \rightarrow 1 \rightarrow 0.6$ RIN Customized Halving interpretation?

No candidate is preferred or adopted before public review.

9.2 What Could Change the Principal Review Set

The principal review set and its interpretation should be reconsidered if:

1. network participants do not consider the fast binary-halving cadence a problem;
2. bounded issuance is not considered a required policy property;
3. credible evidence shows that four-year 50% reward shocks are reliably absorbed by price or miner economics;
4. implementation, arithmetic, or security defects are found in S1, S5/b or S6/b;
5. another bounded scenario performs better under the same published criteria;
or
6. corrected source data materially change the current empirical conclusions;
or
7. the authors or maintainers of Customized Halving materially clarify, revise, withdraw or supersede the S6/b direction.

9.3 Questions for Public Review

1. Should Rincoin retain S0 or change the subsidy rule at height 840,000?
2. Should Rincoin eliminate all large scheduled reward cliffs through S1?
3. Should it retain four-year halving milestones through S5/b?
4. Should it use the bounded S6/b $4 \rightarrow 2 \rightarrow 1 \rightarrow 0.6$ RIN path and exact 168M maximum scheduled issuance?
5. Is S6/b's 36% activation reduction acceptable, and is its multi-century final phase useful or operationally too similar to tail emission?

6. Should the fixed S6/b rule remain separate from the broader governance contingency and the unbounded S7 implementation snapshot?
7. How much value should be assigned to the psychological, communication, and community-coordination role of halving events?
8. Is the approximately 3.6-year delay before S5/b's first future halving a sufficient opportunity for the market and mining ecosystem to mature?
9. Is S1's smoother but less eventful monetary path preferable?
10. Does any evidence support or contradict an expectation that halving milestones improve price, attention, or adoption?
11. Are all three candidate specifications and vector sets technically accurate?
12. Are any eligibility gates, criteria, sources, or calculations incorrect?
13. Should S5/a remain a secondary comparator or enter the principal set?
14. What evidence should change the principal review set?

Maintainers and authors of externally published proposals are specifically invited to submit factual corrections. Where practical, distinguish factual corrections from policy preferences, but free-form feedback is welcome.

10 Implementation, Activation and Public Review

10.1 Implementation and Activation Risks

Any post-840,000 subsidy change is a hard fork. It does not preserve consensus-code continuity after activation. It can preserve ledger continuity if all pre-activation blocks and UTXOs remain valid and only future subsidy validation changes.

Before activation, the project needs:

- final consensus-change specification;
- exact integer arithmetic and rounding rule;
- activation boundary tests;
- transition and off-by-one vectors;
- independent simulator or implementation cross-check;
- deployed-source citations for current consensus behaviour;
- reproducible review-implementation test artifacts;
- wallet, pool, explorer and exchange readiness;
- chain-identification and replay-risk decision;
- go/no-go checkpoint date and height before block 840,000.

If the final specification, implementation, normative test vectors, public review, reproducible releases, pool/exchange/wallet readiness and replay-risk decision are not complete by a defined checkpoint date and checkpoint height, activation at block height 840,000 should be deferred.

Public review should be based on a reproducible package rather than on the PDF alone. The package should include this report, all three candidate whitepapers, all three candidate specifications, Quarto sources, machine-readable scenario configuration, simulation code, all three vector sets, generated data, checksums, immutable source references, public and private feedback routes, and explicit conditions that could change the principal review set.

10.1.1 Chain Identification, Peer Separation and Replay Protection

Changing the subsidy rule changes block validity after activation, but it does not automatically solve transaction replay or service confusion. A block-version marker can identify block-validation semantics if it is explicitly enforced, but it does not by itself prevent the same signed transaction from being valid on two branches that share transaction formats and signature rules.

The replay and service-confusion analysis must cover P2P protocol version, user-agent strings, network magic, default ports, block validation rules, checkpoints, transaction serialization, signature hashing, address formats, wallet chain selection, SPV server selection, exchange routing, explorer labeling and one-sided versus two-sided replay protection.

No final replay-protection mechanism is prescribed here. It requires source-level analysis and a separate technical design.

10.2 Risk Register

Risk	Affected scenarios	Consequence	Mitigation required
Off-by-one activation bug	All hard-fork scenarios	Invalid chain split at block height 840,000	Boundary vectors, regression tests and independent implementation review
Rounding mismatch	S1, S2, S4, both S5 variants, S6 and S7	Different nodes calculate different subsidy	Integer-only specification and normative vectors
Treating the documented 168M maximum as a required target	S2, S4 and S6	Multi-century issuance or coefficient choice without explicit policy	Decide whether it is a ceiling, target, or documentation-versus-consensus inconsistency
Unbounded issuance	S3 and S7	No fixed maximum scheduled issuance	Explicitly decide whether tail emission is acceptable
Replay or service confusion	All hard-fork scenarios	User, wallet and exchange loss	Separate chain-identification and replay analysis
Weak source citations	All scenarios	Publication cannot be independently verified	Immutable source references before final release
Mining-security overclaim	All scenarios	Misleading security expectations	Keep subsidy analysis separate from price/hashrate forecasts
Dated or incomplete empirical evidence	All scenarios	Point-in-time rental, liquidity or orphan-index data may be overgeneralized	Publish retrieval times and hashes; refresh through a separately reviewed evidence snapshot
Unverified rental delivery and hardware efficiency	All scenarios	Ask capacity or normalized cost may be mistaken for executable attack capacity	Controlled benign rental test and reproducible RinHash hardware benchmarks

10.3 Normalized Sensitivity Appendix

Category	Case	Parameter	Value	Unit	Status
price path	Stress	annual price-index change / liquidity realization	-5.0% / 70.0%	normalized sensitivity	illustrative, not forecast
price path	Flat	annual price-index change / liquidity realization	0.0% / 90.0%	normalized sensitivity	illustrative, not forecast
price path	Moderate growth	annual price-index change / liquidity realization	3.0% / 80.0%	normalized sensitivity	illustrative, not forecast
fees	Observed low fees	fee per block / annual change; basis: 90-day all-block median	0 base units / 0.0%	base units per block	derived from frozen observed chain data
fees	Observed base fees	fee per block / annual change; basis: 90-day all-block arithmetic mean, rounded to nearest base unit	10003 base units / 0.0%	base units per block	derived from frozen observed chain data
fees	Observed active-block high fees	fee per block / annual change; basis: 90-day p90 conditional on blocks containing non-coinbase transactions	290698 base units / 0.0%	base units per block	derived from frozen observed chain data; conditional distribution
hardware	Normalized reference hardware	electricity + other operations + amortization	0.40 + 0.10 + 0.20	normalized value per hashrate-index unit-year	placeholder class, not measured RinHash hardware
hardware	Normalized efficient hardware	electricity + other operations + amortization	0.28 + 0.08 + 0.22	normalized value per hashrate-index unit-year	placeholder class, not measured RinHash hardware
hardware	Normalized high-cost hardware	electricity + other operations + amortization	0.55 + 0.15 + 0.20	normalized value per hashrate-index unit-year	placeholder class, not measured RinHash hardware
miner response	Slow	revenue elasticity / response lag	0.50 / 30 days	normalized sensitivity	illustrative, not behavioral law
miner response	Base	revenue elasticity / response lag	1.00 / 14 days	normalized sensitivity	illustrative, not behavioral law
miner response	Fast	revenue elasticity / response lag	1.50 / 3 days	normalized sensitivity	illustrative, not behavioral law

The price paths, liquidity realization, placeholder hardware costs and miner-response elasticities are normalized sensitivities. The fee rows are anchored to the frozen 90-day chain distribution, but the long-run model holds or changes them only as configured. `data/miner_revenue_break_even.csv` and `data/sustainable_hashrate_index.csv` contain the full Cartesian output.

`data/attack_cost_proxies.csv` retains the earlier generic 10%, 25% and 50% redirected-hash shares as secondary stress cases. The empirical rental model in `data/rental_attack_capacity.csv` is primary because it uses visible market depth and current network hashrate. Neither dataset is a complete attack-cost model.

10.4 Generated Artifacts

Artifact	Purpose
data/scenario_config.json	Machine-readable source of truth for simulation parameters
scripts/simulate_monetary_scenarios.py	Scenario generator, assertions, figure writer and include-fragment writer
scripts/collect_empirical_snapshot.py	Collector that produced the reused 365-day base chain snapshot
scripts/collect_transaction_fees.py	Exact prevout-minus-output fee collector with per-block cross-checks
scripts/collect_orphan_evidence.py	Paginated collector for the explorer orphan index
scripts/collect_market_evidence.py	Separate live collector for price, exchange, rental, event and supplemental chain evidence
scripts/collect_transaction_metrics.py	Adds transaction serialization metrics while preserving the exact fee reconstruction
scripts/collect_exchange_observation.py	Append-only daily collector for the planned 14-30 day exchange observation series
scripts/analyze_empirical_snapshot.py	Offline empirical analyzer; performs no live API calls
scripts/analyze_market_evidence.py	Offline market, halving-event and miner-behaviour analyzer
evidence/network-and-security-snapshot-20260727T121555Z/	Frozen 365-day node, explorer, rental and exchange evidence
evidence/market-and-chain-snapshot-20260727T163500Z/	Frozen daily price, market, timeline, halving and transaction-metric evidence
data/empirical_source_manifest.csv	Source URLs, retrieval times and frozen-response hashes
data/market_source_manifest.csv	Frozen market-extension source URLs, retrieval times and response hashes
data/empirical_network_snapshot.csv	Timestamped current-network snapshot

Artifact	Purpose
<code>data/historical_network_baseline.csv</code>	30-, 90- and 365-day network, transaction, fee and indexed-orphan statistics
<code>data/hashrate_history.csv.gz</code>	Per-block difficulty and 120-block rolling hashrate history
<code>data/fee_per_block_history.csv.gz</code>	Per-block subsidy, total fees, transaction count and fee share
<code>data/orphan_history.csv</code>	Frozen explorer-indexed orphan records
<code>data/rental_market_listings.csv</code>	Every visible RinHash rental listing and ask
<code>data/rental_cost_reconciliation.csv</code>	Listing-level ask units, duration minima and reconciled all-capacity prices
<code>data/rental_snapshot_comparison.csv</code>	Timestamped comparison of the network-aligned and later same-day RinHash rental snapshots
<code>data/rental_cost_curve.csv</code>	Capacity-weighted ask curve and contractual minimum costs
<code>data/rental_attack_capacity.csv</code>	Observed-rental capacity against honest-hash and owned-hash stress cases
<code>data/rental_overlap_model.csv</code>	Rental-redirection cases for overlap fractions from 0% through 100%
<code>data/rental_overlap_confirmation_proxies.csv</code>	Mathematical catch-up proxies by overlap case and confirmation depth
<code>data/service_confirmation_policies.csv</code>	Operator-supplied wallet, exchange and atomic-swap settings
<code>data/fee_replacement_calibrated.csv</code>	Boundary fee requirements calibrated to observed fee distributions
<code>data/exchange_liquidity.csv</code>	Venue-separated order-book depth and miner-sale-pressure calculations
<code>data/hardware_benchmarks.csv</code>	Explicitly pending empirical hardware dataset
<code>data/hardware_benchmark_protocol.csv</code>	Reproducible RinHash benchmark protocol
<code>data/empirical_scenario_security_comparison.csv</code>	Scenario comparison using the frozen baseline plus normalized reference values

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Artifact	Purpose
data/rin_daily_price_volume.csv	CoinPaprika RIN/USD, derived RIN/BTC, volume, gaps and confidence flags
data/exact_halving_events.csv	Exact timestamps and subsidy transitions at blocks 210,000, 420,000 and 630,000
data/halving_price_event_windows.csv	One-, seven-, 30- and 90-day price windows without interpolation
data/halving_price_hashrate_event_study.csv	Chain, fee, revenue and confidence metrics around each halving
data/price_compensation.csv	Required price-only compensation at every deterministic reward boundary
data/market_event_timeline.csv	Separately classified chain, project and operator-supplied market events
data/nestex_lp_snapshot.json	Public and operator-supplied NestEx LP snapshots with label distinction
data/nestex_lp_concentration.csv	Provider shares, HHI and effective-provider count
data/nestex_trade_history.csv	Frozen executed trades aggregated from 60 rate-limited pages
data/sticky_elastic_hashrate_scenarios.csv	Committed/elastic miner-exit and rental-plus-owned-hash stress cases
data/transaction_value_size_summary.csv	Economic-value, serialization, input/output and fee-rate distributions
data/transaction_size_by_structure.csv	Serialized-size distributions for the most common exact input/output structures
data/transaction_size_by_value_range.csv	Size and structure summaries by aggregate-output-value range, with value caveats
data/transaction_size_relationships.csv	Descriptive correlation diagnostics; no causal interpretation
data/activation_terminal_shock_comparison.csv	Activation shock, later shock, cadence and terminal regime by scenario
data/pow_security_literature.csv	Bounded primary-source literature review and transfer limits

Artifact	Purpose
includes/*.qmd	Generated QMD table fragments included by this document
data/scenario_summary.csv	Per-scenario deterministic summary
data/horizon_table.csv	Horizon outputs for all scenarios
data/boundary_vectors.csv	Boundary probes for transition tests
data/S1_normative_test_vectors.csv and .json	Candidate S1 normative vectors generated from the same scenario source
data/S1_normative_test_vectors.sha256	Hashes of the generated S1 vector files
data/S5B_normative_test_vectors.csv and .json	Candidate S5/b normative vectors generated from the same scenario source
data/S5B_normative_test_vectors.sha256	Hashes of the generated S5/b vector files
data/S6B_normative_test_vectors.csv and .json	Candidate S6/b boundary, cumulative-issuance and claim-validity vectors
data/S6B_normative_test_vectors.sha256	Hashes of the generated S6/b vector files
data/S6B_independent_verification.json	Results from the independent S6/b calculation that does not import the primary generator
data/S1_S5B_candidate_comparison.csv	Direct timing, cadence, issuance, and terminal comparison
data/S1_S5B_equal_issuance.csv	Ideal and integer equal-total-issuance derivation
data/reward_shock_metrics.csv	Normal scheduled reductions, economically relevant thresholds, and terminal effects
data/s1_subsidy_half_life.csv	Descriptive S1 mathematical half-life
data/threshold_crossings.csv	Below-threshold and zero-subsidy crossings
data/scarcity_dilution_comparison.csv	Direct scarcity and dilution comparison
data/security_assumptions.csv	Normalized economic assumptions and evidence status
data/reward_boundary_shocks.csv	Every post-activation subsidy boundary and nominal revenue shock
data/fee_replacement_requirements.csv	Fee increase required to offset each subsidy reduction

Artifact	Purpose
<code>data/miner_revenue_break_even.csv</code>	Miner-revenue and break-even sensitivity outputs
<code>data/sustainable_hashrate_index.csv</code>	Normalized revenue-supported hashrate-cost index
<code>data/attack_cost_proxies.csv</code>	Redirected-hash and owned-hardware attack-cost proxies
<code>data/security_comparison_by_horizon.csv</code>	Common-assumption scenario comparison
<code>scripts/simulate_security_economics.py</code>	Security-economics generator and consistency checks
<code>figures/*.png</code>	Generated long-horizon and near-term charts
<code>MANIFEST.sha256</code>	SHA-256 manifest for the current public-review package

11 Source Notes and Evidence

Claim or statement	Classification	Current status	Required final evidence
RinHash construction is BLAKE3 -> Argon2d -> SHA3-256	Original-whitepaper claim and deployed-source fact	Confirmed in the supplied PDF and deployed Core (RinCoin Whitepaper 2025 ; Rin-coin contributors 2026f)	Immutable original-whitepaper archive remains pending
Target block interval is 60 seconds	Original-whitepaper claim and deployed-source fact	Confirmed in the supplied PDF and deployed chain parameters (RinCoin Whitepaper 2025 ; Rin-coin contributors 2026a)	Complete
Initial subsidy is 50 RIN	Original-whitepaper claim and deployed-source fact	Confirmed in the supplied PDF and deployed subsidy function (RinCoin Whitepaper 2025 ; Rin-coin contributors 2026c)	Boundary tests remain required for the candidate implementation
Original 168,000,000 RIN maximum-supply statement	Original-whitepaper claim; documentation/consensus mismatch	The supplied PDF states Max supply: 168 million ; deployed binary-halving consensus approaches approximately 21,000,000 RIN; MAX_MONEY is a value-range constant and its source comment says it is not total money supply (RinCoin Whitepaper 2025 ; Rin-coin contributors 2026e , 2026c)	Immutable original-whitepaper archive remains pending
Deployed binary-halving schedule below 840,000	Deployed consensus fact	Deployed Core uses binary right shift (Rin-coin contributors 2026c)	Independent boundary tests remain required
Transaction and block validation	Deployed source fact	The consolidated Core source entry identifies transaction structure, input/value and script checks, the coinbase subsidy-plus-fee limit, and cumulative-work selection (Rin-coin contributors 2026d)	Complete for the cited snapshot
S1 effective maximum scheduled issuance	Simulation-derived result	Generated as 44,624,993.01750000 RIN	Final simulator hash and independent implementation vectors
S5/a effective maximum scheduled issuance	Simulation-derived result	Generated as 32,812,499.76900000 RIN	Final simulator hash and independent implementation vectors
S5/b effective maximum scheduled issuance	Simulation-derived result	Generated as 44,624,999.76900000 RIN	Final simulator hash and independent implementation vectors
Final spendable or circulating supply	Chain/economic outcome	Not equal to scheduled maximum by definition; may be lower	Chain data and accounting methodology if needed

Claim or statement	Classification	Current status	Required final evidence
Tokino Customized Halving scenarios	Public PDF study snapshot	Tokino v1.6.4, chapter 3, printed pages 10-16, defines three scenarios; this analysis models their Layer-1 subsidy paths and excludes annual loss-rate assumptions that are not consensus-enforceable (Tokino 2026)	PDF SHA-256 5269207ea7e363e8df312ed50c00afc119b43e6fa5d3c717e6a7d8fc9863
Proof of Rinne	Separate long-horizon research architecture	Tokino v1.6.4 chapters 4-7 present a temporal-layer and asset-recirculation concept intended to preserve a security budget under a fixed macro cap. It is acknowledged as an interesting research direction, but it is not a present Layer-1 subsidy schedule and is not scored in this analysis (Tokino 2026).	Normative specification, implementable abandonment criterion, threat model, reference code, test vectors and adoption path
Aevust Core Customized Halving code snapshot	Source-code implementation snapshot	Aevust Core commit <code>f1591e4</code> , <code>src/validation.cpp</code> lines 1271-1290, implements a path corresponding to Tokino Scenario II phase rewards but returns 0.6 RIN indefinitely after height 6,300,000 unless another cap rule exists elsewhere; modelled as S7 (Aevust 2026)	Complete for the inspected snapshot; later source changes require a new versioned scenario
Current network and 30/90/365-day history	Direct observation and chain-data derivation	Frozen local-node snapshot at height 674,074 plus 496,274 main-chain block records; response and output hashes recorded (Rincoin Community Forge 2026a)	Independent reproduction from another synchronized node
Transaction fees	Derived from observed chain data	68,946 non-coinbase transaction fees reconstructed from prevouts; each block sum matched <code>getblockstats totalfee</code> (Rincoin Community Forge 2026a)	Independent parser cross-check
RinHash rental exposure	Directly observed ask market and derived cost curve	The network-aligned MRR snapshot exposed 22 available listings and 10.3463 MH/s; a later same-day snapshot exposed 14 listings and 6.6583 MH/s; no 30-day rentals were reported (Rincoin Community Forge 2026a , 2026b)	Controlled benign rental delivery test and repeated market observations

Claim or statement	Classification	Current status	Required final evidence
RinHash hardware efficiency	Empirical measurement	No sufficient reproducible hashrate-plus-wall-power dataset identified	Execute published benchmark protocol on representative hardware

The supplied original RinCoin whitepaper is packaged as `original/rincoin_whitepaper.pdf`, with SHA-256 `82ff8d10264b54828dabb28bf493cae42bcdcb`. It was published at https://coin.rin.so/rincoin_whitepaper.pdf. The content hash fixes the inspected artifact, an independently maintained copy is [in this repository](#). An immutable archive URL remains a publication TODO. The whitepaper is treated as a historical document, not as current executable consensus (*RinCoin Whitepaper 2025*).

Deployed-behaviour evidence uses `Rin-coin/rincoin commit b52c87778f800dc5f4e2f59c372badbc139f933f`. Chain parameters set the 210,000-block subsidy interval, 60-second target spacing, and DGW activation. Subsidy validation uses binary right shift from `50 * COIN`. `COIN` and `MAX_MONEY` are defined in `amount.h`; the comment explicitly distinguishes `MAX_MONEY` from total supply. *Dark Gravity Wave* and *RinHash* are cited at the inspected commit. RinHash uses Argon2d v1.3, `t_cost = 2`, `m_cost = 64`, and `lanes = 1` (*Rin-coin contributors 2026a, 2026c, 2026e, 2026b, 2026f*).

The base empirical security snapshot is frozen under `evidence/network-and-security-snapshot-20260727T121555Z/`. It contains local-node JSON-RPC results and 365 days of `getblockstats/` header history, exact per-transaction fee reconstruction, explorer summary and orphan-index responses, Mining Rig Rentals listings and market page, NestEx and Rabid Rabbit order books, and a Kraken XBT/USD conversion quote. The local node was fully synchronized at block height 674,074.

The frozen market extension is stored under `evidence/market-and-chain-snapshot-20260727T163500Z/`. It contains CoinPaprika RIN and BTC daily history and the HTTP 402 limitation response, current venue metadata, NestEx order-book, liquidity and 60 tradebook pages, Mining Rig Rentals and Rabid Rabbit snapshots, the explorer’s `getlastorphans` response, exact local-node halving blocks, supplemental pre-halving chain history, transaction serialization metrics and frozen GitHub timeline records. Each `SOURCE_MANIFEST.json` records source URL, UTC retrieval time, byte count and SHA-256 hash. All public market values remain dated observations rather than current facts at publication time (*Rincoin Community Forge 2026a, 2026b*).

The externally published Customized Halving scenarios were read from [Tokino Rincoin v1.6.4](#), chapter 3, printed pages 10-16, SHA-256 5269207ea7e363e8df312ed50c00afc119b43e6fa5d3c717e6a7d8fc9863147b. This analysis treats the study as a source of scenario definitions and research claims, not as deployed consensus ([Tokino 2026](#)).

The same paper develops Proof of Rinne in later chapters as a separate long-horizon regenerative architecture. This report acknowledges that research direction but does not model or score it as a present Layer-1 monetary candidate. Its exclusion is a scope decision: the supplied material does not yet define a consensus-safe proof of permanent abandonment, complete adversarial ownership-recovery rules, deployed reference code, normative vectors or a block-840,000 activation path.

The Aevust implementation evidence is [Aevust/rincoin commit f1591e44277f6acb77d7b3d734da1cc884d05edb](#), dated 2026-05-23. [src/validation.cpp lines 1271-1290](#) implement 4 RIN from height 840,000, 2 RIN from 2,100,000, 1 RIN from 4,200,000, and 0.6 RIN from 6,300,000 onward. The inspected subsidy function does not itself stop issuance at 168 million RIN; unless another consensus rule enforces a terminal height elsewhere, this snapshot is modelled separately as S7, an unbounded implementation variant rather than the finite-to-cap S6/b study scenario ([Aevust 2026](#)).

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